Beyond Lexical Projection: The Role of Little n and the DP Layer in the Linearization and Interpretation of Korean Relative Clauses

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1 Introduction

Unlike English, Korean is a head-final language, where relative clauses (RCs) function as modifiers that precede nouns (Chae, 2012). RC adnominal phrases are marked by four different morphemes: *-eun*, *-neun*, *-eul*, and *-deon* (Hong, 2012), which are suffixed to verbs or adjectives. The onsetless markers *-eul* [ɯl] and *-eun* [ɯn] can fuse with the preceding syllable, in which case only the coda consonants [l] and [n] surface. Notably, verbs and adjectives are the only conjugating lexical classes among the nine parts of speech in Korean (noun, pronoun, numeral, particle/postposition, interjection, verb, adjective, determiner, and adverb). Because conjugation attaches RC markers to the end of the verbal phrase (VP) or adjectival phrase (AdjP), the basic structural format of a Korean relative clause can be represented as follows. In the Figure 1, the onsetless RC marker *-eun* has been fused with the preceding light verb *-ha*, resulting in the surface form *-n*-ADN.

[Kim-uywon-ul	myengyehoysonha-n]	kica	
Kim-senator-ACC	defame-ADN	reporter	(Han, 2013)

Figure 1: An example of a Korean relative clause

The head-finality of Korean, combined with the pre-nominal position of relative clauses and their marking via verbal morphology (in lieu of an overt complementizer), creates derivational challenges from a minimalist perspective. For one, it would require the verbal or adjectival modifier to enter the numeration already bearing relative-clause morphology, which introduces a look-ahead problem. This is problematic because a clause constitutes a spell-out domain (phase; Chomsky and Kenstowicz (1999)), and operations internal to that domain are no longer accessible to the NP head to which the clause later adjoins. Furthermore, proposals addressing the source of double relative clause constructions (DRCs) argue that they arise from double nominative constructions (DNCs) (Han and Kim, 2004). This analysis crucially assumes that some relative clauses originate from a fully formed declarative sentence with a declarative sentence-final ending. For the RC morphology to enter the derivation, there must therefore be a projection higher than the RC+NP maximal

projection—a functional head capable of hosting the relevant morphosyntactic features. The original declarative ending can then be deleted by an independent PF rule.

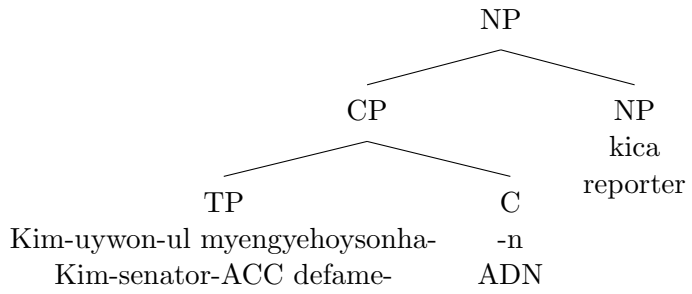


Figure 2: Structure of the Korean relative clause in (1).

A second issue concerns word order. Under the Linear Correspondence Axiom (LCA), linear precedence is determined by the asymmetric c-commanding relationship. However, the hierarchy in Fig. 2 incorrectly predicts that the head noun following the relative clause should be pronounced earlier, contrary to the attested pre-nominal RC order. This linearization problem is also mitigated if we posit an additional functional head in the nominal domain and allow the relevant phrase to raise to the specifier of that projection. Such movement produces the correct c-command relations and yields the surface order in which the relative clause precedes the head noun.

Given that the RC + modified head noun constituent is still distributionally a noun and should project the category of its head (i.e., be endocentric), we posit the functional category to be the little n. The RC marker, then, is hosted at little n. Drawing on parallels between the clausal and nominal domains (Adger, 2003), the relative clause marker plays a conceptual role similar to that of a light verb, in that its meaning is determined only in relation to the modifying relationship specified within the NP. For example, the morphemes *-eun*, *-neun*, *-eul*, and *-deon* are homophonous with other grammatical functions, such as the topic markers *-eun*, *-neun* and the accusative marker *-eul*. When these morphemes appear in the little n position, they will acquire the meaning of a relative clause only in the appropriate context. Moreover, these morphemes encode dual functions: they mark the modifying ending while simultaneously encoding the tense and aspect of the event. Namely, the form and meaning do not exhibit a one-to-one correspondence (Hong, 2012); *-eun* marks [+past] in event predicates but [-past] in stative predicates. Consequently, the predicative information present within the RC is necessary for the RC marker in little n to receive its interpretation—much like the verb *take*, which is interpreted differently in *take a nap* versus *take a bus*.

The postulated syntactic structure illustrated in the tree of Figure 3 provides two additional benefits. First, it allows the coindexation of the operator and the head noun to be established locally, as in a sisterhood configuration. Under the dominant operator-movement analysis of Korean relative clauses (Han, 2013), the head noun is formed external to the relative clause, while the operator undergoes a A' movement. An issue with this approach is that the dependency between the operator

and the head noun cannot be licensed in narrow syntax and must instead be supplied at LF through semantic variable binding. In the proposed structure, after the relative clause is copied from its base position as an adjunct to the NP head and merged into spec nP, a scattered deletion operation may apply: in the lower copy, the RC material deletes, leaving only a CP skeleton containing the operator in spec CP, while in the higher copy, the operator deletes as the overt RC material remains. The residual CP in the lower copy can then be reanalyzed as an independent argument, placing the operator in a head–complement configuration with the head noun. This structural relation licenses the dependency syntactically, eliminating the need for LF-level variable binding.

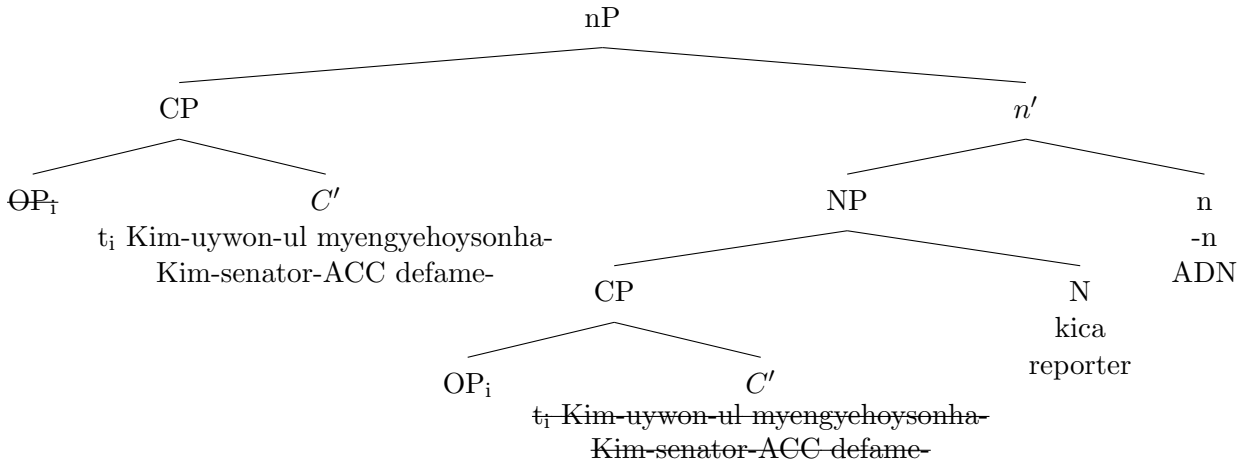


Figure 3: Proposed nP projection of 1.

Second, this approach also allows us to account for the possessive relationship between the head noun and its modifying predicates, which is tacitly assumed in stand-alone relative clause constructions. Korean relative clauses can be classified into two types: (1) the stand-alone type, in which a single verb or adjective independently modifies the noun, with the gap syntactically corresponding to the subject position, while the head noun semantically interpreted as possessing the property expressed by the elided subject; and (2) the phrase type, in which a verb phrase containing either an object (for subject gaps) or a subject (for object gaps) modifies the head noun. In Figure 4b, the head noun "book" corresponds to the object position of the verb phrase with the omitted object (i.e., "Cheolsu friend likes"), consistent with a Korean SOV word order.

[____ keu-n] saram	[Cheolsu-ga ____ joaha-neun] chingu
big-ADN person	Cheolsu-NOM like-ADN friend
(a) Stand-alone type	(b) Phrase type (object gap)

Figure 4: Relative Clause Types in Korean

Meanwhile, in Figure 4a, the non-relativized sentence may literally correspond to "the person is big," but it can also be interpreted as "the person's height, size, or head is big" (Chae, 2012). This latter interpretation assumes that the property in question (e.g., height, size, or head) is the

elided subject, as illustrated in Figure 5a. Because Korean is a pro-drop language, such elision is permissible; in these cases, the genitive/possessive marker *-uy*, which normally precedes the subject to establish the modifying relationship between the head noun and its property, is omitted alongside the elided subject. Both elements would be restored when reconstructing the non-relativized version of the sentence (Figure 5b). Note that *-i* and *-ga* are allomorphs of the Korean nominative case marker, appearing after consonant-final and vowel-final stems, respectively.

[____ (ki-ga/momjip-i/meori-ga) keu-n] saram (height/size/head)-NOM big-ADN person	saram-uy ki-ga/momjip-i/meori-ga keu-ta person-GEN height/size/head-NOM big-DECL
(a) Stand-alone RCs with omitted subjects	(b) Reconstruction of possessive relations

Figure 5: Stand-alone relative clauses and their possessive reconstruction

Mapping this construction into a syntactic tree, the elided noun functions as the agent of the descriptive property. We may therefore hypothesize that it is tacitly merged in spec nP—phonologically null due to elision but semantically present for interpretation—following an extension of the Universal Theta-role Assignment Hypothesis (UTAH) to the nominal domain (Adger, 2003). Meanwhile, the head noun, as the possessor of the property, is copied and merged into spec DP to receive genitive case during reconstruction. The corresponding tree is provided in Figure 6. Note that this reconstruction involves covert movement at LF, motivated solely by interpretive considerations, with no phonological effect. That is, the PF form (Figure 5a) remains unchanged.

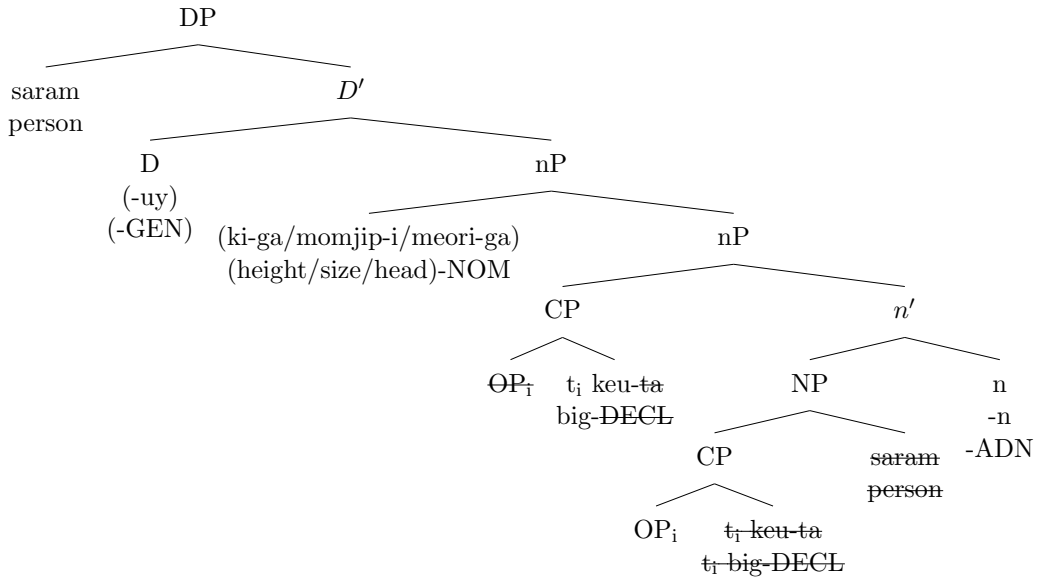


Figure 6: Reconstruction tree of 5b.

Two additional points warrant attention from Figure 6. First, as the relative clause marker hosted at little n merges as the complement of NP, the declarative marker originally generated at the base position—and subsequently moved to spec nP—is deleted. As noted earlier, this deletion is triggered by an independent PF rule, presumably to prevent two word endings from being affixed

to a single verb or adjective stem. Although the illustration concerns operations at LF, the deletion is included here for expository purposes. More importantly, once the head noun moves to spec DP, its local syntactic binding relation with the operator is disrupted, thereby forfeiting the advantage of local coindexation discussed earlier.

We may mitigate the severity of this issue when considering the nature of the possessive interpretation in the stand-alone construction (as opposed to the literal interpretation in Figure 4a, where the individual is merely interpreted as being big, without specification of the relevant property). Under the possessive interpretation, the head noun no longer functions as the direct agent corresponding to the gap; rather, its semantic role is demoted to that of a modifier standing in a possessive relation to the elided subject. Consequently, the need to establish a strictly syntactic binding relation becomes less pressing, unlike in phrase-type RCs (Figure 4b) and in stand-alone RCs under the literal interpretation, where the head noun directly corresponds to the gap.

Thus far, we have identified two motivations for positing a functional head in the Korean relative clause construction: the problems posed by inflecting morphology in the numeration and by word order. Although this functional head, little n, is an established concept in the broader literature (e.g. Adger (2003)), it has not previously been implemented in analyses of relative clauses. The proposed superimposed X-bar structure provides a suitable host for the relative clause marker, allowing verbal and adjectival morphology to be declarative without encountering a look-ahead problem at numeration. At the same time, it supplies a specifier position to which the relative clause can raise, thereby preserving the pre-nominal word order via an asymmetric c-command relation in accordance with the LCA. In addition, we have shown that this account yields two interpretive benefits: one bringing variable binding into the syntactic domain, and the other accounting for the possessive interpretation of the stand-alone construction.

Given this overview, the remainder of the paper is structured as follows. In section 2, we review the dominant operator-movement analysis of Han (2013) and show how it gives rise to the two problems identified above. In section 3, we introduce our proposal and demonstrate how it addresses each of these limitations while also offering additional theoretical advantages.

2 The Operator-Movement Analysis and Its Limitations

2.1 The Operator-Movement Approach

Unlike in English, the head-raising analysis does not hold for Korean relative clause constructions. Diagnostics based on the constituency of idioms, adjectival modifiers, and anaphors indicate that the head noun cannot have been base-generated within the relative clause (Han, 2013). In particular, (i) an object–verb idiom loses its idiomatic interpretation when the object is relativized out of a declarative construction; (ii) an adjective adjacent to the head noun can only receive a high-attachment interpretation to the matrix predicate, not to the embedded verb inside the relative clause; and (iii) in sentences where the antecedent of a head noun anaphor is inside the relative

clause, the anaphor cannot be properly bound.

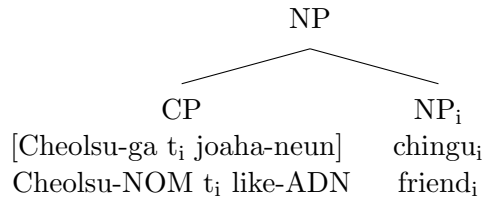
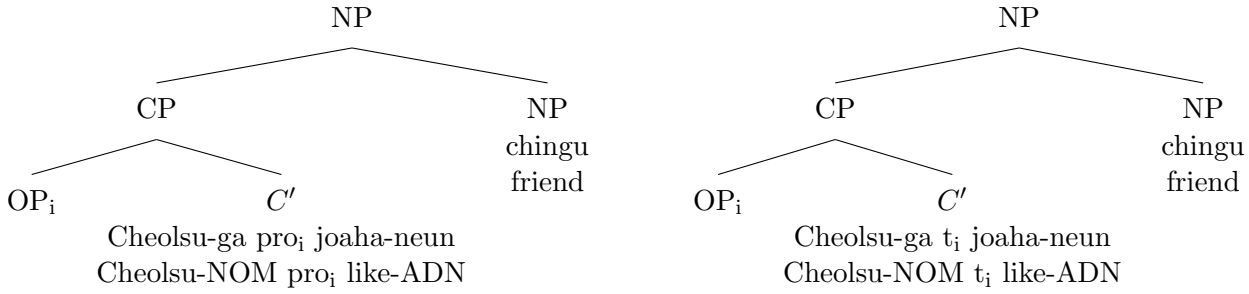


Figure 7: Head-raising analysis of 4b.

In the head-raising approach, generating the modifying element phrase-internally offers a syntactic advantage in that the raised nominal can derivationally bind its trace (or copy). Figure 7 shows how the head-raising analysis operates for the example in Figure 4b. As can be seen, because the relationship between the gap and the head NP is explicitly represented in the syntax, there is no need to resort to a syntax-external abstraction.

Given that this expository advantage conflicts with the empirical data in Korean, analyses must instead posit an operator that either moves out of the relative clause (the operator-movement analysis) or binds a null pronoun *in situ* (the operator-binding analysis), while the modified noun itself is externally merged. Among the two approaches, empirical evidence favors the operator-movement analysis (Kang, 1988; Kwon, 2008), as the gap is sensitive to acceptability judgments in island constructions, whereas the corresponding position does not allow overt pronominal realization except in certain embedded contexts—a pattern unexpected if the null element were simply bound *in situ*.



(a) Operator-binding analysis

(b) Operator-movement analysis

Figure 8: RC analysis assuming an externally merged head

Unlike in the head-raising approach, the operator trace or null pronoun in these analyses must be semantically linked to the modifying entity via an independent mechanism of coindexation. Figure 8 presents two analyses of the same sentence shown in Figure 7 from these perspectives. As can be seen, the head noun that is interpretively associated with the gap—filled by trace t_i in the operator-movement analysis or by the null pronoun pro_i in the operator-binding analysis—does not enter into an explicit syntactic indexation relation with it. While this constitutes a structural draw-

back of the approach, it is not a direct focus of the present discussion; rather, it will be resolved as a consequence of addressing the more fundamental issues associated with conceptualizing the head as an externally merged entity. Moreover, although the operator-movement approach will be taken as the default analysis in what follows, the problem at issue arises equally for both approaches that posit a non-lexical operator. For this reason, the discussion will proceed with both approaches in mind, though the exposition will target the movement analysis more specifically.

Importantly, these operator-related difficulties are merely surface symptoms of a deeper problem inherent to external merge itself. This underlying problem becomes clear once we consider how the relative clause is assembled. Since the head noun is not structurally contained within the relative clause, the adjunct relative clause, having been independently constructed prior to adjunction, lacks access to the nominal it will ultimately modify, making advance specification of their dependency impossible. Chapter 8 of Hornstein et al. (2005) notes that adjuncts are not required to accompany the head noun throughout the derivation and may instead be formed independently, with the point at which they are merged left flexible. This assumption has been used to explain the acceptability of sentences such as *Which claim that John$_i$ made did he$_i$ discuss?* Since the relative clause *that John made* can be merged after wh-movement has taken place, the sentence does not incur a violation of Principle B. Extending the same logic: if relative clauses containing a null operator (e.g., *Cheolsu-ga OP$_i$ joaha-neun*) were constructed prior to adjunction, how could we ensure that the verb surfaces with the relative clause marker *-neun* rather than, say, the declarative ending *-ta*?

2.2 Problem 1: Numeration and Inflectional Morphology

Having noted why pre-specification of relative clause morphology is problematic under externally merged-head approaches, we can now illustrate the issue more concretely by examining how the numeration would have to be assembled in such systems. Following Chomsky (2015)'s notation, as adopted in Hornstein et al. (2005), the lexical items present in the numeration for the example in Figure 4b can be represented as follows.

N = { Cheolsu-ga$_1$, joaha-neun$_1$, chingu$_1$ }^{1}

(a) Numeration for head-raising

Glosses: *Cheolsu-ga* = Cheolsu-NOM; *joaha-neun* = like-ADN; *chingu* = friend

N$_1$ = { Cheolsu-ga$_1$, joaha-ta$_1$, OP/pro$_1$ }, N$_2$ = { chingu$_1$ }^{2}

(b) Numeration for external merging

Glosses: *Cheolsu-ga* = Cheolsu-NOM; *joaha-ta* = like-DECL; *chingu* = friend

Figure 9: Numeration under head-raising vs. external-merge analyses

Under the external-merge analysis (Figure 9b), two independently assembled numerations must be postulated—one for the relative clause (N_1) and one for the head-containing matrix structure (N_2). In contrast, the head-raising analysis (Figure 9a) requires only a single numeration, since the head noun originates within the relative clause and thus belongs to the same derivational workspace. Importantly, in the minimalist framework adopted by Hornstein et al. (2005), lexical

items are introduced into the numeration with their inflectional features already specified—a property consistent with the elimination of a distinct S-structure level and the assumption that case features are checked only at LF. Subsequently, the phrases in Figures 7 and 8 are derived via Selection and Merge, as follows:

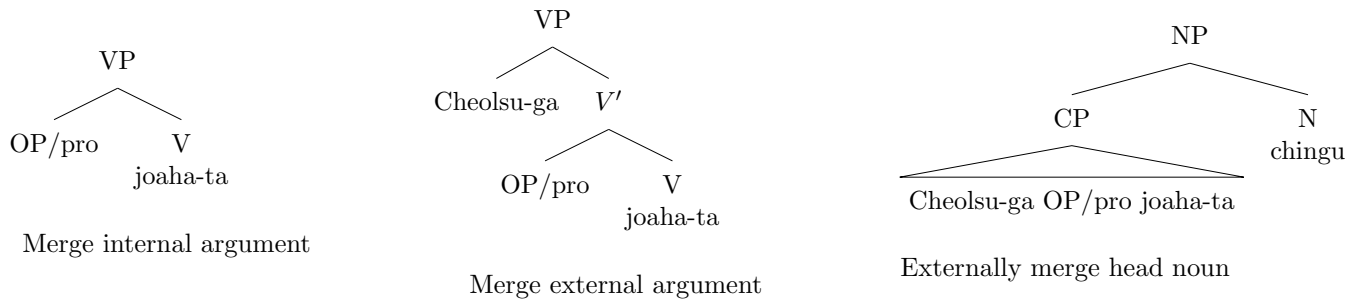


Figure 10: Deriving Relative Clauses with Externally Merged Heads

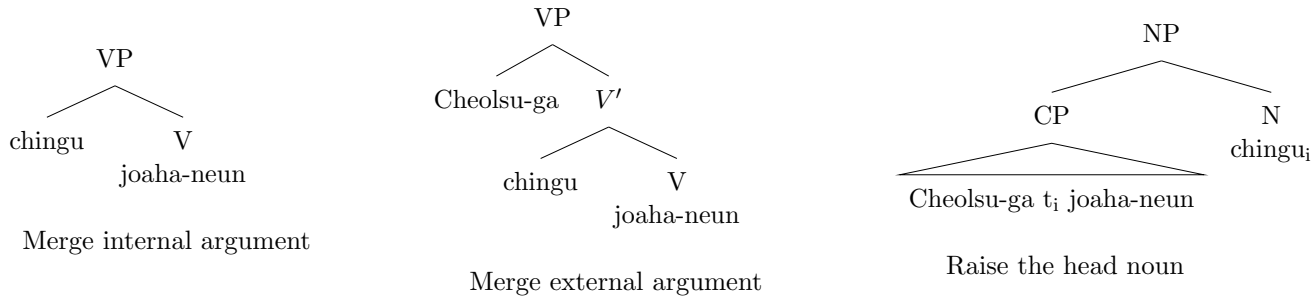


Figure 11: Deriving Relative Clauses with Internally Raised Head

In Figure 10, because the adnominal phrase is built from a separate numeration, the verb surfaces with the declarative ending *-ta*, rather than with the required adnominal ending *-neun*. In contrast, in the head-raising derivation in Figure 11, all relevant elements originate within a single, homogeneous numeration sensitive to the intended use of the verb, allowing the correct adnominal morphology to be generated. This contrast becomes clearer when we examine the expanded representations of the rightmost structures from Figure 10 and Figure 11, shown in Figure 12.

Figure 12: Derivational Outcomes Under Different Numeration Assumptions

Another perspective on this issue comes from the view that CP constitutes a phase, a cyclic

domain relevant for spell-out. As proposed in Chomsky and Kenstowicz (1999), spell-out proceeds cyclically, thereby limiting the domain accessible to syntactic operations at any given point. This reduction of the active workspace constrains the amount of material that must be inspected during syntactic computation and thus eases the burden on working memory. Crucially, the theory posits the Phase-Impenetrability Condition (PIC), which states that the internal domain of a phase—the complement of the phase head—becomes inaccessible to operations initiated outside that phase. This is because the phase complement undergoes spell-out and is transferred to the phonological interface, placing it beyond the reach of narrow syntax. Consequently, even if the TP inside a declarative CP (i.e., Figure 12a) required further RC-licensing to surface with the adnominal marker, the PIC renders such licensing impossible, since the spelled-out TP is no longer available for syntactic operations.

Perhaps one may circumvent the difficulty of hosting an appropriate verbal ending by appealing to look-ahead in the numeration and pre-specifying the relative-clause morphology in the adjunct numeration (Figure 9b). Indeed, the operator-binding approach may resolve the ending-morpheme conflict, provided that some flexibility in the derivational ordering is permitted. Nevertheless, even this possibility fails to rescue the operator-movement analysis from its clause-typing problem, given its commitment to the strict derivational timeline assumed for double relative clauses (DRCs). This commitment is necessary for the analysis to remain consistent with the seemingly unfavorable evidence posed by DRCs—namely, that the operator associated with a relativized subject appears to move out of another relative clause, in violation of the island constraint.

[RC1	[RC2	e_i	e_j	joaha-neun]	chingu$_j$-ga	yeepu-n]	Cheolsu$_i$
				like-ADN	friend$_j$-NOM	pretty-ADN	Cheolsu$_i$

Figure 13: Example of Double Relative Clause (DRC)

The double relative clause in Figure 13, adapted from the example in Figure 4b, illustrates a case in which a subject (Cheolsu$_i$) inside an already relativized CP (RC2) undergoes further relativization, resulting in movement that violates island constraints. Although the bracketed representation here does not involve a head-raising analysis, the gap is coindexed with the noun entities to make the intended semantic connection explicit. The full-fledged mapping of the standard analysis, which assumes genuine double relativization, is provided in Figure 14. Under this analysis, the operator associated with t_i should, in principle, be blocked from moving out of RC1 (CP) by an island constraint or a phase boundary. Yet the construction appears to permit exactly such movement, which—if it were truly illicit—would instead force t_i to be interpreted as a null pronoun bound *in situ* rather than as the lower copy of an operator chain.

To circumvent this evidence that conflicts with the operator-movement analysis, Han and Kim (2004) propose that DRC examples arise from an independently well-formed structure—namely, the double nominative construction (DNC). The DNC in Korean is characterized by sentences

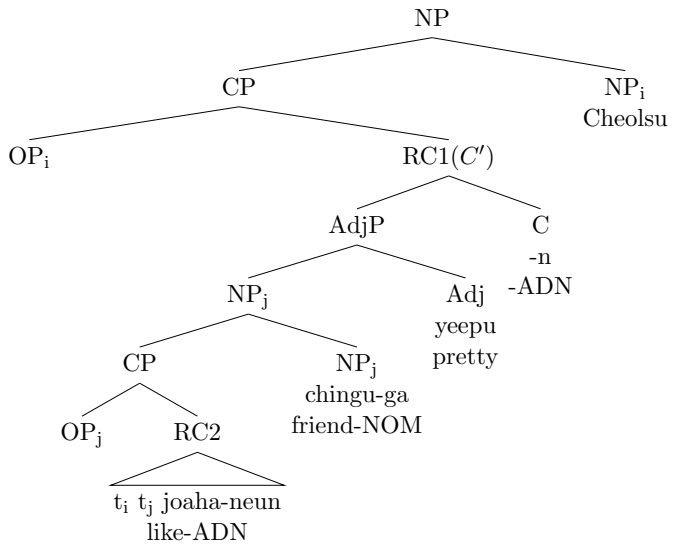
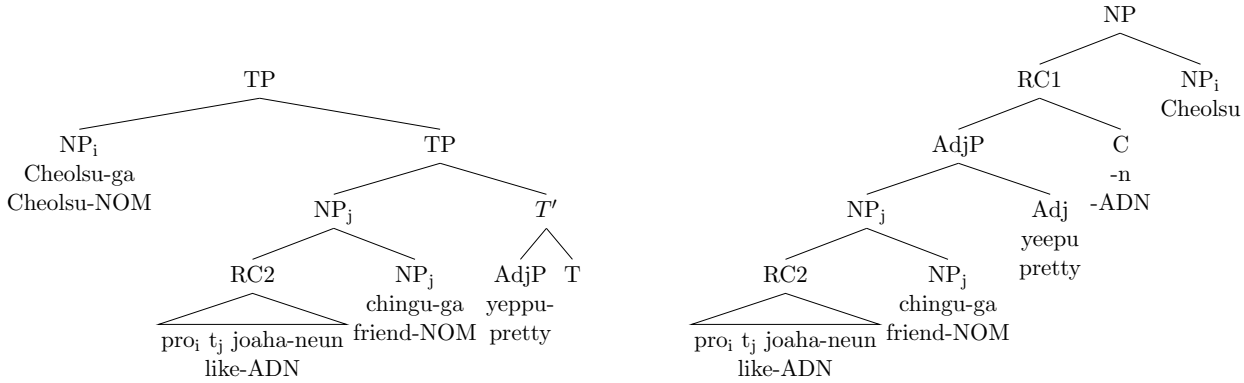


Figure 14: Standard Analysis of Figure 13

with multiple subjects, where the first nominative is adjoined to the IP (here, TP) formed by the second nominative and the predicate (YOON, 1986). In Figure 15a, one can observe that the first nominative NP_i adjoins to the TP formed by the second nominative NP_j and the predicate AdjP. When the second nominative forming the internal IP (TP) is modified by a relative clause containing a null pronoun coindexed with the first nominative, what appears to be double relativization can instead be derived through a single instance of relativization of the first nominative subject.



(a) DNC starting point

(b) Relativization of NP_i

Figure 15: DNC-Based Analysis according to Han and Kim (2004)

Figure 15b follows this proposed logic: the internal RC2 is inherited from the DNC starting point represented in Figure 15a. Crucially, because RC2 contains the null pronoun pro_i rather than the operator trace t_i found in Figure 14, the resulting relativization does not incur an island violation, as no operator movement is involved. The authors posit that every DRC instance is derived from its corresponding DNC counterpart. For the example at hand, Figure 13 corresponds to Figure 16, which presents a flattened, bracketed simplification of the tree in Figure 15a.

Cheolsu$_i$-ga	[RC2 pro$_i$	e$_j$	joaha-neun]	chingu$_j$-ga	yeepu-ta
Cheolsu$_i$-NOM			like-ADN	friend$_j$-NOM	pretty-DECL

Figure 16: Double Nominative Construction corresponding to the DRC in Figure 13

What this analysis fails to capture, however, is the operator movement that would have been required for the embedded relative clause *RC2* if the account were to remain consistent with the operator-movement approach it aims to defend. That is, while the null pronoun pro_i in Figure 16 does not require a separate movement step, the operator associated with e_j would. This movement would generate a CP layer above the proposed double-nominative IP (TP), whose complementizer position—the slot ordinarily filled by a conjugational ending—would be realized with the declarative marker. Figure 17 draws out this expanded version of the simplified source structure in Figure 15a. The occupancy of C by a declarative ending is also consistent with the DNC examples presented in Han and Kim (2004), all of which surface with the final *-ta* (as in Figure 16).

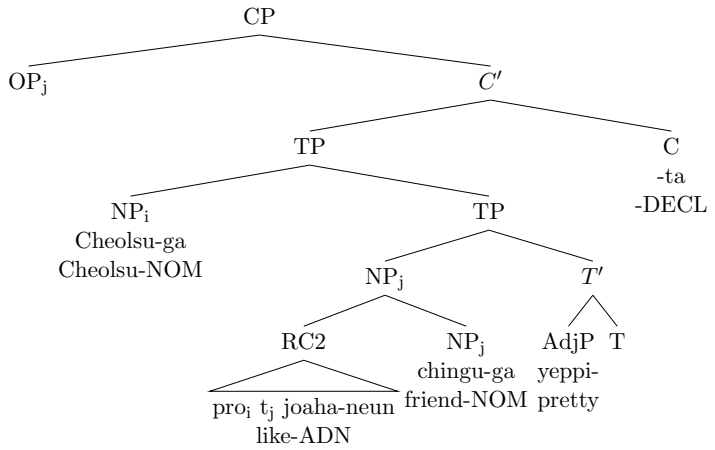


Figure 17: Expanded Tree Corresponding to the DNC Structure in Figure 15a

Granted that there is no independent mechanism for supplying the relative-clause morpheme when the first nominative NP$_i$ (Cheolsu) is relativized, we are faced with the same external-merge problem discussed earlier. Importantly, however, this problem cannot be avoided by appealing to a lenient notion of look-ahead in the numeration, since the analysis explicitly treats the constituents of the double-nominative configuration—including an adjectival predicate with a declarative ending—as a fixed lexical derivation point. As a result, the derivation affords no stage at which the appropriate relative-clause morphology could be introduced, and it crashes when the verb is forced to surface with the declarative ending instead (Figure 18).

2.3 Problem 2: Word Order

Chapter 6 of Hornstein et al. (2005) identifies three foundational tasks in building phrases: merging, labeling, and imposing a linear order. While the previous section identified a merging-related problem in the operator-movement analysis of relative clause construction—specifically, the absence of

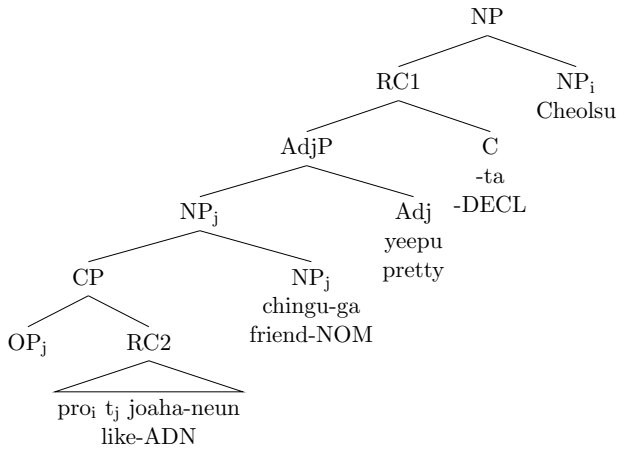


Figure 18: Expanded Tree Corresponding to the Relativized Structure in Figure 15b

a structural position to host the relevant morphological ending—this account also encounters a difficulty with linearization. Structurally, relative clauses belong to the class of adjuncts, a component whose positional status remains a vexed and largely unresolved problem in generative grammar. Adjuncts can adjoin on either side of a binary-branching structure, regardless of the headedness of the language; their licensing is therefore non-directional, and their distribution exhibits a mobile-like flexibility around the head (V, T/I, or C) that licenses them.

Bowers (1993) addresses the apparent contradiction between non-directional licensing and the seemingly asymmetrical distribution exhibited by certain manner adverbs. He uses this observation to motivate the Predicate Phrase (PredP), which creates a configuration where an adverb licensed by V head—which subsequently raises to Pred—cannot precede the verb in the surface word order. If PredP is one device for capturing the asymmetry observed with low adverbs, then some analogous account is required for the word order of adjunct relative clauses, since these too constitute a class of modifiers whose placement cannot simply be left to non-directional licensing. A pre-nominal relative clause cannot surface post-nominally, nor can a post-nominal one surface pre-nominally.

The Linear Correspondence Axiom (LCA) (Kayne, 1994) provides a mechanism for translating syntactic relations into linear order by mapping asymmetric c-command onto a precedence relation. Asymmetric c-command obtains when one constituent c-commands another, but not vice versa, such that the two do not stand in a relation of mutual c-command; hence the asymmetry. Taking the adapted structure from Bowers (1993) discussed above, Figure 19 illustrates how lexical precedence is derived from syntactic structure. Because the verb *learn*, having raised to the head of PredP, asymmetrically c-commands the adverb *perfectly*—a member of the class of manner adverbs that cannot precede the verb in surface order—the adverb must follow the verb. In contrast,*quickly*, which is likewise a manner adverb but belongs to the PredP-licensed subclass, may precede the verb when adjoined from the left branch. Note that the trace, or deleted lower copy, is not considered

in this computation: since the LCA is a PF rule, phonologically invisible material is excluded from the linearization procedure, an important tenet of the theory.

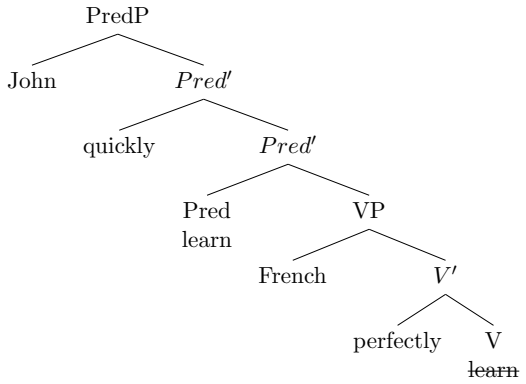


Figure 19: LCA-derived precedence relations, adapted from Bowers (1993)

Importantly, asymmetric c-command is computed only over maximal projections and terminal nodes (heads). This restriction arises, in part, from the cognitive assumption that phonology is sensitive only to information associated with individual lexical items or fully formed phrases. On the practical side, limiting the computation to terminals and maximal projections prevents the system from yielding an indecisive state where virtually every constituent stands in a mutual c-command relation with another. This is possible because, under the LCA, all material contained within a maximal projection inherits the c-command profile of that projection—effectively "piggybacking" on the projection's linearization.

Intermediate projections are disregarded in this enterprise, a fact that renders operator-based analyses (Figures 8a and 8b) more problematic than the head-raising counterpart (Figure 7). In the former, once the operator occupies spec CP, the relative clause itself is relegated to the C' intermediate projection. Because LCA does not treat constituents inside C' as having the same linearization status as the externally merged head noun, the head noun instead asymmetrically c-commands the relative clause, thereby incorrectly yielding a post-nominal word order. The head-raising analysis (Figure 7) is not exempt from word-ordering concerns either, although the problem is somewhat less severe: in this case, the relative clause—corresponding to the maximal projection CP—stands in a mutual c-command relation with the head noun. Still, for all three approaches, a mechanism is required to ensure that the relative clause asymmetrically c-commands the head noun if the phonological system is to "read" the pre-nominal word order. Hornstein et al. (2005) cites one way to rescue the mutual c-commanding relationship is through movement: by moving the element that must precede in the morphological ordering into a position where it asymmetrically c-commands its counterpart. This is the stance the present paper builds its solution on.

Lastly, it should be noted that the current formulation of the LCA lacks a theoretical consensus on how adjuncts, including relative clauses, are to be linearized. By drawing on the intuition behind

the theory, the present analysis seeks to fill this gap in empirical coverage and offer a proposal for phenomena that existing approaches have only partially addressed.

3 Our Proposal

So far, we have identified two structural deficiencies that our revision must address. First, the derivation requires a position in which the relative-clause marker can be introduced. Second, the analysis must provide a configuration that allows the relative clause to raise and asymmetrically c-command the head noun so that the pre-nominal word order can be yielded. We propose that introducing a functional head above the maximal projection formed by the relative clause and its modified head noun (as in Figures 7 and 8) provides a unified solution to these issues.

Nevertheless, we must be aware that the modified NP still distributionally functions as a noun—that is, as a nominal argument. This follows from the endocentricity of the structure: in both Figures 7 and 8, the projection is headed by the noun, which serves as the structurally prominent element within the phrase. Thus, the functional category must remain within the nominal domain so as not to alter the categorial status of the phrase. Related insight can be taken from a line of argument that suggests the topmost projection of the nominal domain to be the determiner phrase instead of NP (Adger, 2003). Although this view is debated for article-less languages like Korean, as it presupposes a theoretically posited but phonologically less active functional head—a move against the minimalist spirit—the difficulty that current theoretical devices face in explaining the derivation and word order of Korean relative clauses suggests that such an additional layer of projection may nonetheless offer explanatory value.

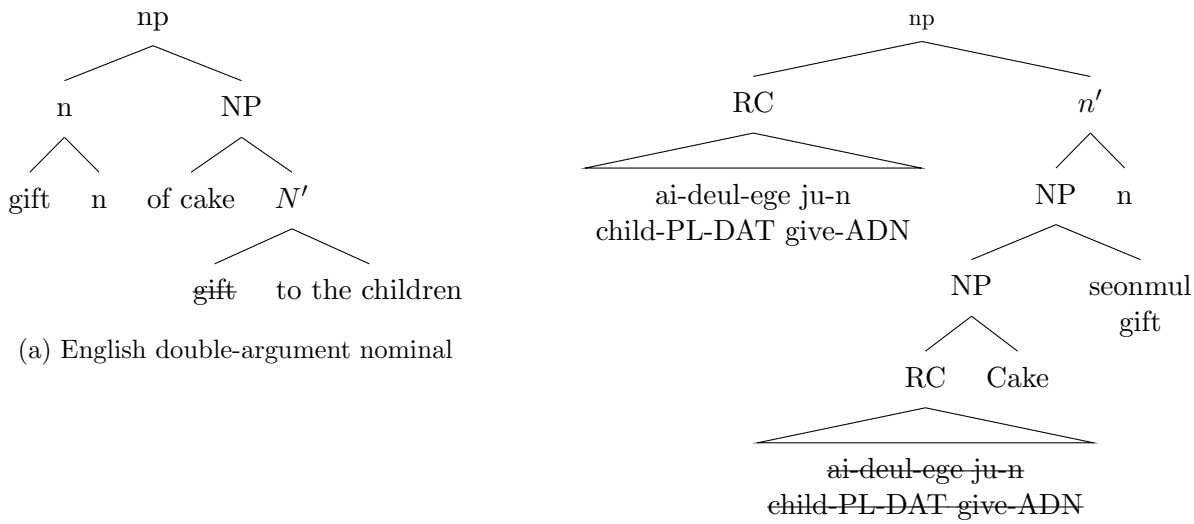
What makes this theory particularly appealing for our purposes is that it concerns non–theta-related semantics while offering mechanisms to handle movement and phrase ordering. These two properties map directly onto the challenges identified above: on the one hand, we require a configuration that allows the semantics of the relative clause, signified by its ending morphemes, to properly enter the modifying construction; on the other hand, we need a structural position for the adnominal phrase that satisfies the precedence relations imposed by the LCA.

Notably, this expanded theory of the nominal domain draws on the parallel between how arguments of nouns are licensed within the DP and how arguments of verbs are licensed within the clausal domain, thereby establishing a hierarchy of projections analogous to that of the clause. In the nominal domain, the D head serves as the final closing head, analogous to T, while an additional functional layer, little n, intervenes between D and the lexical category N—much like little v occupying the position between T and the lexical category V.

Since the framework was motivated by observations on the distributional properties of DP arguments within the noun phrase, its baseline analysis centers on argument-taking nouns. These mainly consist of classes of deverbal nouns, such as *gift*, *destruction*, and *analysis*. The theta-role–assigning

configuration mirrors that of verbal phrases: the complement of the lexical category N is the position where the theme is merged (or spec NP, in cases where both a theme and a goal are present, similar to the double-shell structure in ditransitive constructions). spec nP is where the agent argument is introduced, which eventually raises to spec DP to receive genitive case—just as the subject merged in spec vP raises to spec TP to receive nominative case.

As we attempt a cross-linguistic application of these claims, it is important to clarify how the syntax of the nominal domain applies to Korean. Adger (2003) notes that cross-linguistic variation in word order within DP is determined by the strength of the features that trigger movement. In the English data that the theory primarily addresses, (i) little n is projected regardless of whether an agent is merged in spec nP, for raising N to n preserves the word order in double-argument structures—the noun precedes both the theme and the goal in examples such as *the gift of cake to the children*. By a similar token, we may posit that the projection of little n is necessary in Korean in order to provide a specifier position for the relative clause to raise into. This conceptualization becomes even more relevant when we consider that Korean translations of double-argument nominals typically surface with the goal–verb sequence realized as a relative clause. Taking the previous example sentence (Figure 20a), Figure 20b illustrates how it translates into Korean, where the structure surfaces as *[goal + verb-ADN] + theme + head noun*. Such a configuration requires raising in order to maintain linearity under the LCA.



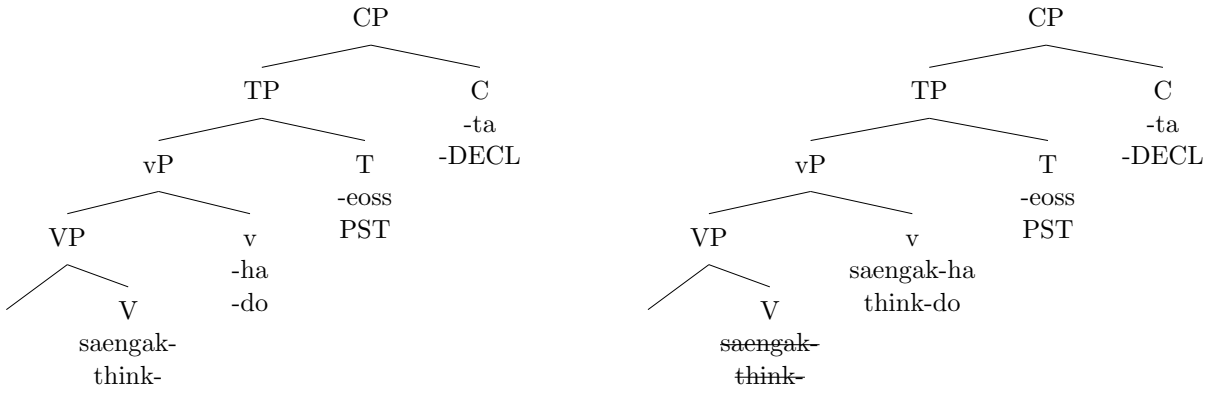
(a) English double-argument nominal

(b) Korean: [goal + verb-ADN] + theme + N

Figure 20: Motivation for projecting little n to maintain the word order

Hidden in claim (i) and illustrated in Figure 20a is the second consideration that follows from the cross-linguistic discussion of featural strength: English noun phrase arguments undergo N-to-n movement. This is triggered by the strong [N] feature on little n, parallel to the strong [V] feature on little v. In contrast, Korean grammar does not require V-to-v movement. The movement is

rather optional, indicating that the v head lacks a strong, uninterpretable [V] feature that would otherwise force overt verb raising. This follows from a morphosyntactic property of Korean: as a head-final, agglutinative language, the heads V, v, T, and C do not appear as independently ordered projections but instead surface as a single fused morphological complex. As shown in Figure 21, the form *saengak-ha-eoss-ta* ('thought') appears as a single morphological unit.



(a) Without V to v Movement

(b) With V to v Movement

Figure 21: Agglutinative V–v–T–C structure in Korean

Consequently, overt V-to-v movement makes no observable contribution to surface word order, as the lexical verb stem and its functional head ultimately appear as an adjoined string regardless (see Figure 21b). Given the weak [V] feature of the light verb, it is reasonable to extend this trait to the nominal domain and conjecture that little n likewise lacks a strong [N] feature. Ultimately, there is no need to assume N-to-n movement in the analysis of Korean.

With N-to-n movement absent, the lexical noun does not occupy little n, allowing n to serve as the locus for auxiliary features that encode non–theta-related nominal semantics. We propose that the relative clause ending morphology in Korean is one such feature, given its distributional properties that parallel those of light verbs. In particular, the interpretation of a light verb depends heavily on the VP complement it selects. Take the verb run: depending on its complement, it can mean 'to manage' (*run a company*) or 'to execute' (*run a program*). Given the parallel between the clausal and nominal domains, the interpretation of little n should likewise depend on its NP complement. Even if little n appears semantically underspecified or polysemous on the surface, its semantic contribution can be determined by the contextual information provided by the lexical projection. This pattern closely mirrors the behavior of relative clause morphology in Korean.

Hong (2012) notes that, contrary to what is expected in an agglutinative language, Korean relative clause markers do not maintain a strict one-to-one correspondence between form and function. That is, the distinct functional morphemes used in relative clause formation do not map exhaustively to a unique grammatical function. This issue arises in three respects. First, there are four

markers corresponding to the single grammatical function of relative clause marking: *-eun*, *-neun*, *-eul*, and *-deon*. Second, these markers not only signal the modificational nature of the phrase but also encode tense and aspect. Third, when these two dimensions of complexity intersect, some morphemes come to express different tense features ([±past]) depending on whether they appear with eventive or stative predicates, and, conversely, the same tense feature may be realized by different morphemes across predicate types.
The author attempts to resolve these mismatches through morphological reanalysis, treating the surface forms as fused exponents of multiple features. The event-predicate RC marker *–neun*, for instance, is analyzed as a combination of a zero [–past] morpheme, the progressive morpheme *–neun*, and the RC marker proper *–eun*. This approach, however, falls short because it relies on retroactive interpretive stipulations. In particular, Hong adopts a disambiguation strategy whereby, if a form is compatible with both present and past interpretations, and a distinct morpheme exists that unambiguously marks past, then the ambiguous form is interpreted as non-past.

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To put this issue into perspective, the fact of the matter is that the interpretation of a relative clause marker hinges on the contextual information of the phrase it suffixes to. This follows naturally if we assume that the RC marker is hosted on little n, which is merged above the relative clause–head NP complex, rather than being inherently pre-specified as part of the conjugational morphology inside the relative clause itself. Under this structural view, the apparent form–function mismatches do not arise from lexical ambiguity that requires retroactive interpretive fixes. Rather, they emerge from the way little n composes semantic meaning by referencing the tense features of the T head within the modifying relative clause contained in its complement. Our proposal—that the RC marker originates in little n—thus captures the observed patterns derivationally, without resorting to stipulation.

This expository advantage is illustrated in Figure 22, which reanalyzes how the semantics of the event-predicate RC marker *-eun* are determined within our derivational framework. The example sentence itself is taken from Hong (2012), whose interpretation originally depended on retroactive stipulation. Note that the relative clause has also moved out of its base-merged position into spec nP, as shown in Figure 20b, which we presented when arguing that little n invariably projects in

Korean.

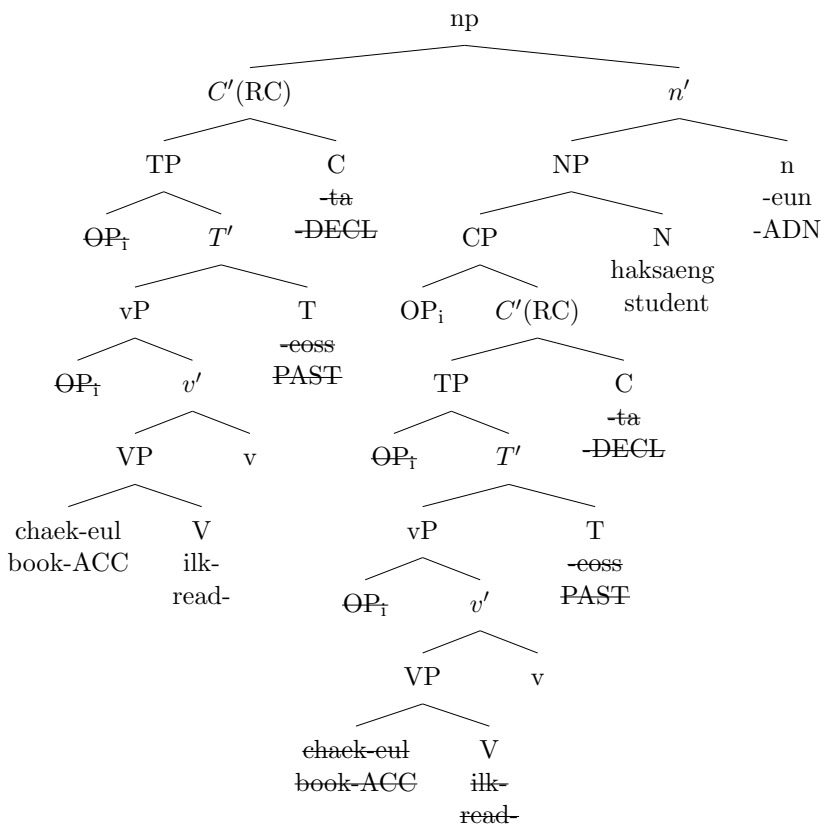


Figure 22: Reanalyzing the RC Marker from a Derivational Perspective

As the tree structure is linearized under the LCA, the full sentence preserves the morphological ordering of *[chaek-eul ilk-eun] haeksaeng* 'the student who read a book.' Notice that *–eun* is introduced by little n, while the original C head—the declarative ending *–ta*—is deleted, which we assume for now to result from a PF rule that prevents two conjugational endings from surfacing simultaneously. Importantly, the tense value of *–eun* is past, as it is determined by the T head *–eoss* 'PAST' inside the relative clause. On the surface, *–eun* receives a past interpretation with event predicates but a present interpretation with state predicates. Rather than disambiguating this contrast retroactively—for example, by reasoning that because event predicate already bears the present-tense morpheme *–neun*, *–eun* must denote past—we can derive the distinction directly from the T head contained within the complement of *n*.

Lastly, the past-tense morpheme *–eoss* is also deleted due to a separate PF rule that disallows *–eoss* from surfacing together with the relative clause marker *–eun*. We do not address this issue further, but note that it does not weaken our analysis, as this assumption is already made in the original account.

This derivational approach offers another advantage that the morphological reanalysis framework struggles to capture. It allows us to account for the tense feature of the form *–eul*, which remains a conundrum under the fused-morpheme account that treats each RC marker as a composite exponent of tense/aspect and the "proper" RC marker *–eun*. In this context, *–eul* becomes problematic, since it lacks any phonological resemblance to *–eun*—unlike *–neun* or *–deon*—that would motivate such morphemic decomposition. However, if we instead assume that *–eul* is a morphemically independent entity whose semantics are structurally supplied by the context of the relative clause, we are spared the burden of accounting for this theoretical loophole.

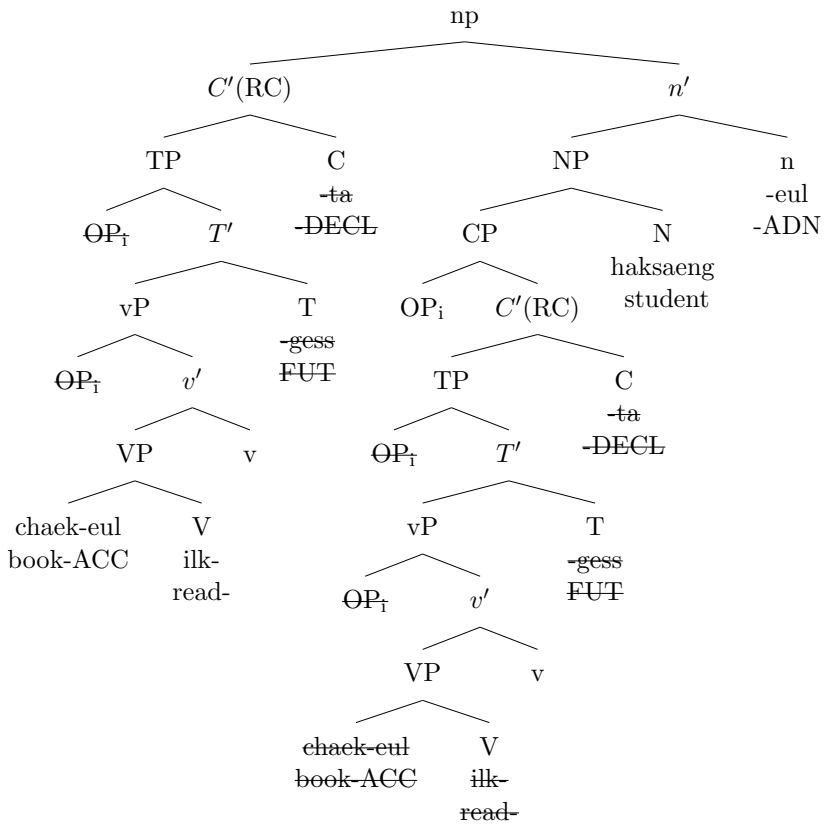


Figure 23: Derivational Account of the Semantics of the *–eul* Morpheme

The tree in Figure 23, whose linear precedence reads as *[chaek-eul ilk-eul] haeksaeng* 'the student who will read a book' (as determined by the relevant asymmetrical c-commanding relationship), illustrates how the future tense associated with the relative clause morpheme *–eul* is transmitted from its dependency on its complement—namely, the relative clause in the C' position *chaek-eul ilk-gess-ta* 'will read a book'. That is, the semantics of the RC marker *–eul* depend on the future-tense morpheme *–gess* hosted on the T head from which it receives its interpretive features. Note that the head–complement relation in this context is defined with respect to the base-merged position, of which a lower copy is subsequently deleted.

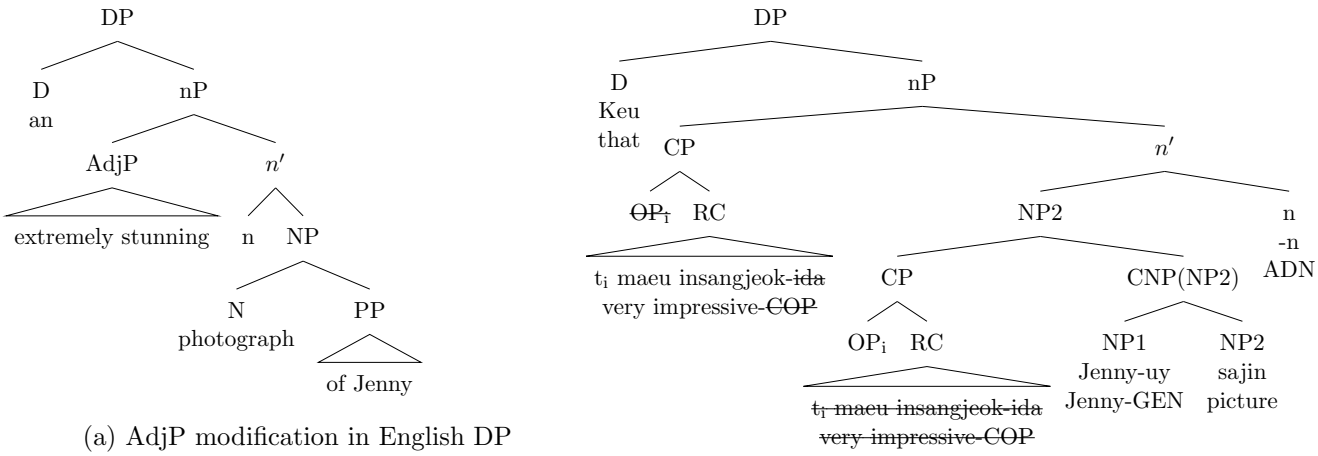
Once again, we encounter a separate PF rule that deletes *–gess* in the environment of the RC marker *–eul*. Although we cannot fully account for this PF constraint here, we justify its plausibility by noting its parallel to the deletion of the past-tense morpheme *–eoss* before the RC marker *–eun*.

The third important parameter to consider is the strength of the genitive case feature on D. In Adger (2003), the [gen] feature of English D is analyzed as strong, enabling it to attract not only the agent base-merged in the specifier of nP but also the theme merged inside NP, analogous to unaccusative subject raising from the VP-complement position. In the latter case, n does not project a specifier position—one in which an agent argument would be generated. This logic attests to the word order whereby, when a relevant argument and a determiner co-occur, the strong genitive feature forces the argument to surface before the determiner (e.g., *Evan's every desire*).

To see how this extends to Korean, we turn to the clausal domain on which the nominal analysis is modeled. In English, T bears an EPP feature that forces subjects to raise, including unaccusative subjects originating within the VP. This same strength is mirrored in the nominal domain, where D strongly attracts a genitive argument—even those originating inside NP. We thus see that the attraction force is consistent across the clausal and nominal domains within the language. Meanwhile, Korean is a pro-drop language that lacks a strong nominal requirement and does not possess an EPP feature. The absence of such strength in the clausal domain predicts that the nominal domain likewise need not attract a genitive subject, as D is expected to bear only a weak genitive feature.
This conjecture aligns with the writer's native-speaker judgment. When translating the earlier English example *Evan's every desire*, which was used to illustrate spec nP to spec DP movement, the Korean expression allows both orders— 'every Evan's desire' and 'Evan's every desire'—although the former is somewhat less natural. The availability of these two orders suggests that Korean D does not enforce the same strong attraction of genitive arguments that is observed in English.

Nonetheless, the attraction of an argument may still occur covertly, as the interpretation of stand-alone relative clauses often involves construing the head noun as the possessor of an elided property (Figure 6). The reconstructed, non-relativized form of such structures presents an interesting case, as the argument within NP is interpreted as raised while the agent—covertly base-generated in spec nP—remains in situ. We return to this issue in detail in section 3.3. For now, it suffices to note that, although the original analysis constrains the coexistence of the possessor and the agent, this syntactic constraint dissolves in the interpretation of relative clauses. It is also important to emphasize that the head nouns under consideration are largely non-deverbal and therefore do not independently bear argument structure; for such nouns, case features are not directly relevant apart from their role in reconstruction. However, this does not undermine the present analysis, as n is invariably projected regardless of whether the noun itself encodes an argument structure.

Lastly, Adger (2003) discusses the potential merging position of adjectival phrase modifiers, a point highly relevant to our discussion of the positionality of RC adjuncts within the nominal domain. The examples they provide situate the modifying phrase between the D and the N head, as in the sentence *an [extremely stunning] photograph of Jenny* (Figure 24a). This leads to the conjecture that AdjP modifiers adjoin to the maximal projection of little n, which structurally intervenes between N and D in the projectional hierarchy (D > n > N).



(a) AdjP modification in English DP

(b) Stand-alone RC modification in Korean

Figure 24: Position of modifying Phrases in DP

Interestingly, Korean relative clause data exhibit the same ordering pattern, following the D head–RC–N head sequence. This similarity is relevant because some analyses (e.g. Chae (2012)) interpret Korean relative clauses as adjectival participial phrases, drawing on evidence that the gap assumed to be a necessary component of such structures behaves like a null anaphor and may, in fact, be optional. Without delving into the details of this analytical perspective, one can nonetheless conjecture the adjectival nature of Korean relative clauses from the existence of stand-alone forms (Figure 4a), which can be interchangeably interpreted as an adjectival modification (i.e., *big person*).

In this vein, the adjective phrase *extremely stunning* can be translated as a standalone relative clause that modifies the NP *Jenny's picture* pre-nominally. In Figure 24b, the relative clause is located between the noun phrase (NP2) *Jenny-uy sajin* and the demonstrative D head (used here because Korean lacks a grammatical article and only optionally employs a demonstrative). Note that although Korean is generally a head-final language, D is left-headed. Additionally, the translated sentence induces an attachment ambiguity whereby the relative clause can modify either the NP1 (Jenny) or the NP2 (picture). Given the high attachment preference in Korean, we have represented the structure accordingly, with the relative clause modifying the complex noun phrase (CNP) headed by NP2 (picture).

The cross-linguistic similarity in word order provides further evidence that the RC adnominal

phrase must ultimately be located in spec nP. Since Korean relative clauses are base-merged inside the lexical layer NP, we can conjecture that they bear the strong [n] feature that motivates their movement to the functional projection. It is more plausible to locate this movement-triggering feature on the RC rather than on n, since adjuncts are optional and not all NPs contain a relative clause that could satisfy such a requirement.

So far, we have established several properties of the Korean nominal domain that fall outside the predictions of Adger (2003). First, nP is invariably projected, a fact that aligns with and helps maintain the word order. Second, D bears a weak genitive feature—one that does not force overt raising but may nonetheless give rise to covert movement in interpretation.Third, n does not bear a strong N feature, as there is no obligatory N-to-n movement; with n left vacant of lexical material, treating it as analogous to v allows it to function as the host for relative clause markers. Fourth, relative clauses make use of a strong n feature that triggers movement to spec nP. Finally, the position occupied by the RC as a result of this movement is cross-linguistically attested, paralleling the English case in which an adjectival modifier occupies a comparable structural site.

All of these observations must be understood in light of the fact that the head nouns in our data—the lexical items around which the relevant functional projections are built—are largely non-argumental. With these tenets in place, the following sections return to the issues raised in section 2, demonstrating how our proposal resolves the example sentences that were previously considered problematic.

3.1 Addressing the Problem in Numeration

In section 2.2, we argued that one central aspect of our proposal is that the availability of little n as a host for the relative clause marker—as a lexical item that independently enters the numeration—allows us to incorporate the modifying ending into the conjugational morphology without requiring it to be pre-specified in an adjunct numeration (N_1 in Figure 9b). As a result, this approach avoids a look-ahead problem, while also preventing cases in which no appropriate ending is available, thereby causing the derivation to crash (Figure 12a).

We further showed that this inclusiveness problem is even more fatal for the operator-movement approach than for the operator-binding approach. The former explicitly assumes a derivational starting point involving a double nominative construction with a declarative ending (Figure 15a); if this stage is to remain consistent with the movement mechanism (Figure 8b), the derivation inevitably encounters a configuration in which no suitable host for relative clause morphology is available (see how Figure 17 leads to Figure 18).

To demonstrate how our revision addresses these two issues, we first show how the availability of the projection headed by little n leads to an appropriate merge step that can replace the configurations in Figure 10, thereby resolving the problem illustrated in Figure 12a. We then examine how it applies to the derivation of double relative clauses (DRCs), by complementing the starting

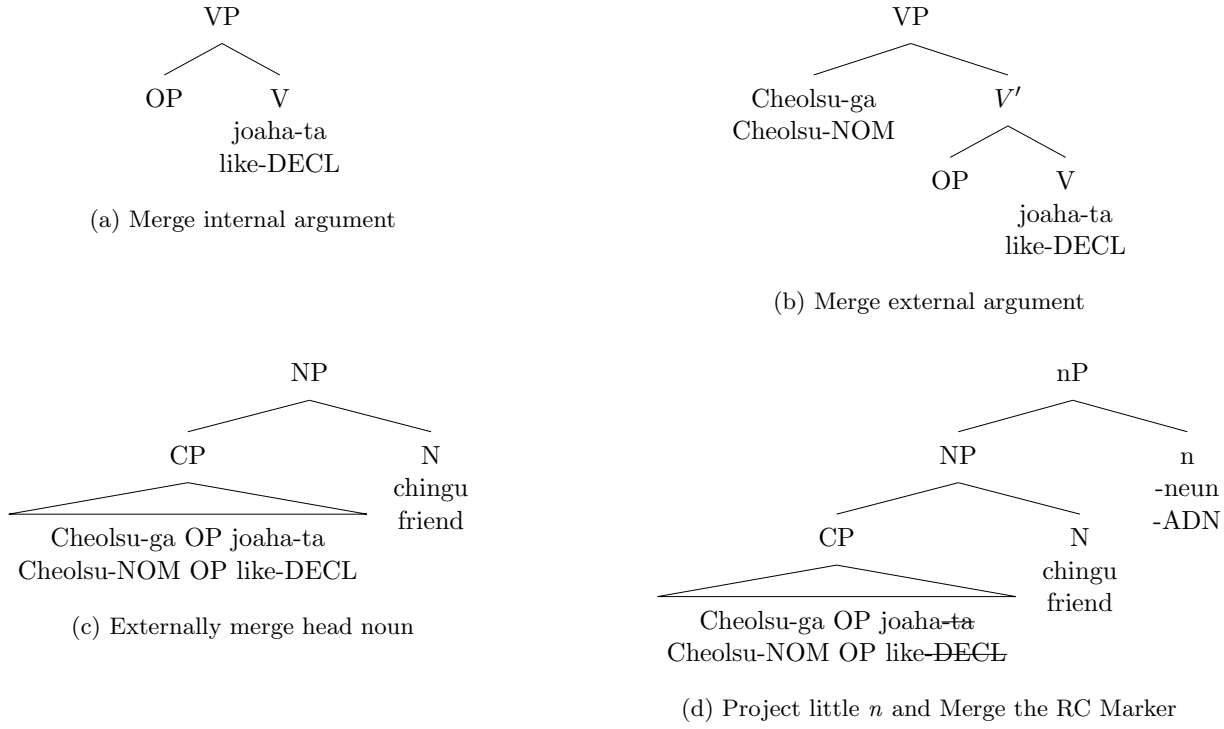
point in Figure 17 so as to account for how the relationship between Figures 15a and 15b can be established without contradicting the assumed operator-movement–based relativization scheme.

$$N_1 = \{ \text{ Cheolsu-ga}_1, \text{ joaha-ta}_1, \text{ OP/pro}_1 \ \},$$
$$N_2 = \{ \text{ chingu}_1, \text{ -neun}_1 \ \}$$

Figure 25: Numeration for external merging assuming nP

Figure 25 illustrates the revised numeration starting point, where the adnominal suffix *-neun* enters the system as an independent lexical item, in contrast to Figure 9b, where it enters as a fused form with the verb *joaha-* 'like', forming *joaha-neun*.

The reason that it is the numeration of the matrix NP (N_2), rather than that of the relative clause adjunct (N_1), selecting *-neun* from the lexicon follows from endocentricity. Recall that the modified NP is distributionally nominal and is headed by the head noun; the modified subject—in our case, *Cheolsu*—is therefore the prominent element responsible for determining the grammatical properties of the projection it heads. In this sense, it is the matrix NP that initiates and regulates the modifying relation. This conclusion is further supported by considerations of cyclicity. Unlike CP relative clauses, which constitute phases and are therefore subject to the Phase Impenetrability Condition (PIC), N is not a phase head. As a result, the N does not trigger spell-out and remains accessible to operations applying beyond its lexical projections, including merger in the nP layer.



(a) Merge internal argument

(b) Merge external argument

(c) Externally merge head noun

(d) Project little n and Merge the RC Marker

Figure 26: Relative Clause Adjunction via little n

Figure 26 revises the merging steps in Figure 10 by introducing little n as the host of *-neun*. At this stage, the asymmetric c-command relation does not yet yield the desired linear precedence: the

structure in Figure 26d would be read off as *neun chingu Cheolsu-ga joaha* (ADN friend Cheolsu-NOM like). This follows from the assumption that copy-and-merge movement of the relative clause phrase takes place only after the initial merger, a derivational step that will be addressed in section 3.2. Meanwhile, the operator also remains in its base position in Figures 26c and 26d. This choice is purely representational, intended to keep the diagram focused on the structural relation relevant to n. Under cyclicity, however, operator movement must apply prior to the spell-out of the complement of the phase head C. Related details beyond this simplified representation will likewise be addressed in the following section.

At this point, we instead introduce the PF constraint responsible for deleting the declarative ending *-ta*. Rather than treating this effect as an artifact of the present analysis, we take it to follow from a theory-neutral constraint on complementary distribution. Namely, human natural language does not allow double occupancy of a single head. For instance, a structure such as *I like envy a friend* is ungrammatical because the verbs *like* and *envy* would occupy the same lexical V head (raised to v in English), even though a single head can host only one lexical item. To encode a parallel conflict in the nominal domain, we require a slight modification that prevents the simultaneous realization of n and C when the relative clause adjoins to N as an adjunct.

Definition 1 (Complementary Distribution of *C* and little *n* under NP-Adjunction)**.** *The C head associated with a modifying phrase base-merged within an NP cannot simultaneously be realized with little n in the same nominal domain.*

This stipulation is also applicable to English, the language on which much of the nominal-domain analysis has been based. In English, adjuncts are adjoined to the maximal projection of n (Figure 24a), unlike Korean relative clauses, which adjoin within the lexical layer (Figure 24b). The former configuration is therefore exempt from the PF constraint proposed above, allowing the relative clause complementizer and n—occupied by a lexical item raised from N—to co-occur within a single DP domain. In this regard, Figure 27 illustrates how *that* and *destruction* are co-realized.

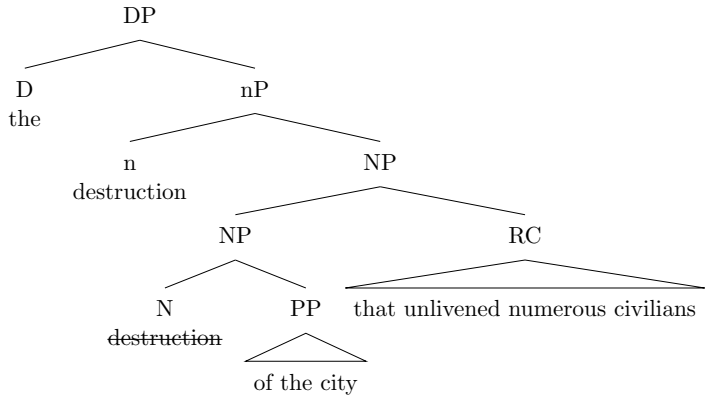


Figure 27: An English structure exempt from the PF rule in Definition 1

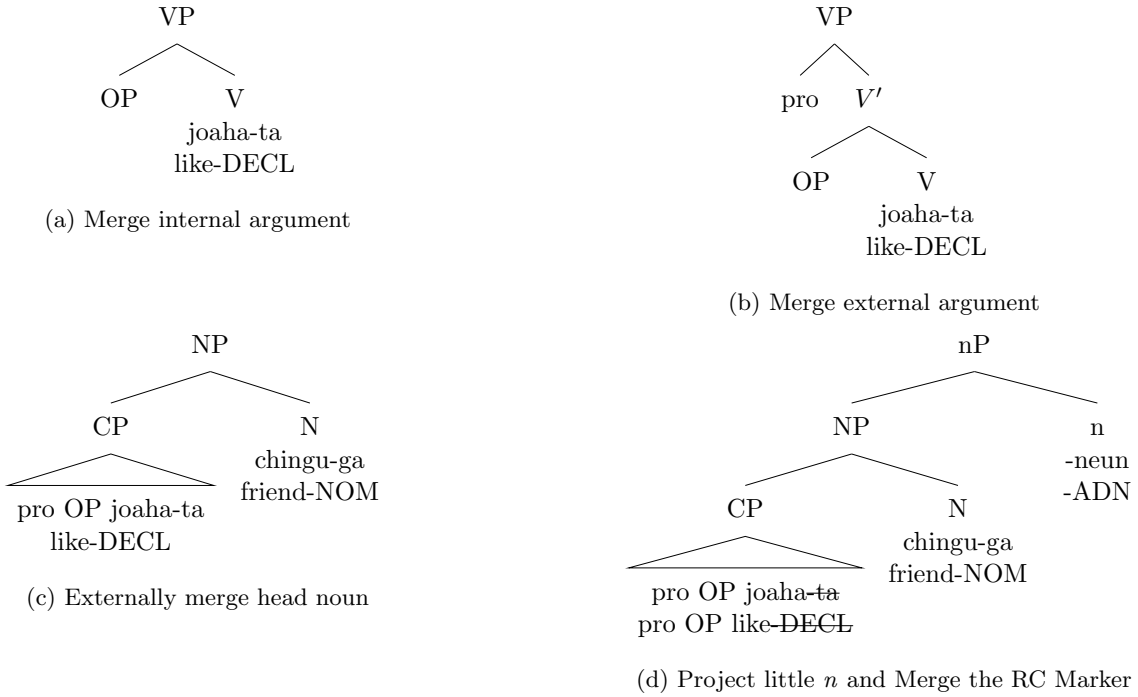
We now turn to the derivation of double relative clauses (DRCs). Following Han and Kim (2004), who analyzes DRCs as a single relativized instance of the double nominative construction (DNC),

we take the lexical elements of the DNC as the derivational starting point for the relative clause. The numeration used for this adjunct modifier stands separate from that of the matrix NP, with which it eventually merges. This follows the logic introduced earlier in section 2.1, where adjuncts are independently constructed in a complete form before adjoining to the modified subject. In Figure 28, N_1 illustrates the adjunct numeration derived from the DNC in Figure 16, while N_2 represents the numeration for the matrix NP.

$$N_1 = \{ \text{pro}_1, \text{OP}_1, \text{joaha-ta}_1, \text{chingu-ga}_1 \text{ yeppu-ta}_1, \text{-neun}_1 \},$$
$$N_2 = \{ \text{Chelosu}_1, \text{-neun}_1 \}$$

Figure 28: Numeration for Double Relative Clauses assuming the DNC starting point

The merging of the selected lexical items in Figure 28 proceeds in two steps. The first step concerns the second nominative NP. Recall that in DNCs, the second nominative is a modified RC+NP constituent that contains a null pronoun coindexed with the first subject (see Figures 15a and 17). While this inner relative clause (RC2) was previously assumed to be simply pre-configured at the starting point, a full account of the derivation requires that it be addressed as well. The formation process largely follows that of simple relative clause adjunction, as introduced in Figure 26, with a variation to the externally merged subject. Because the relative clause modifying the second subject contains a null pronoun rather than a phonetically overt subject, *pro* is this element that is merged at the second step (Figure 29b).



(a) Merge internal argument

(b) Merge external argument

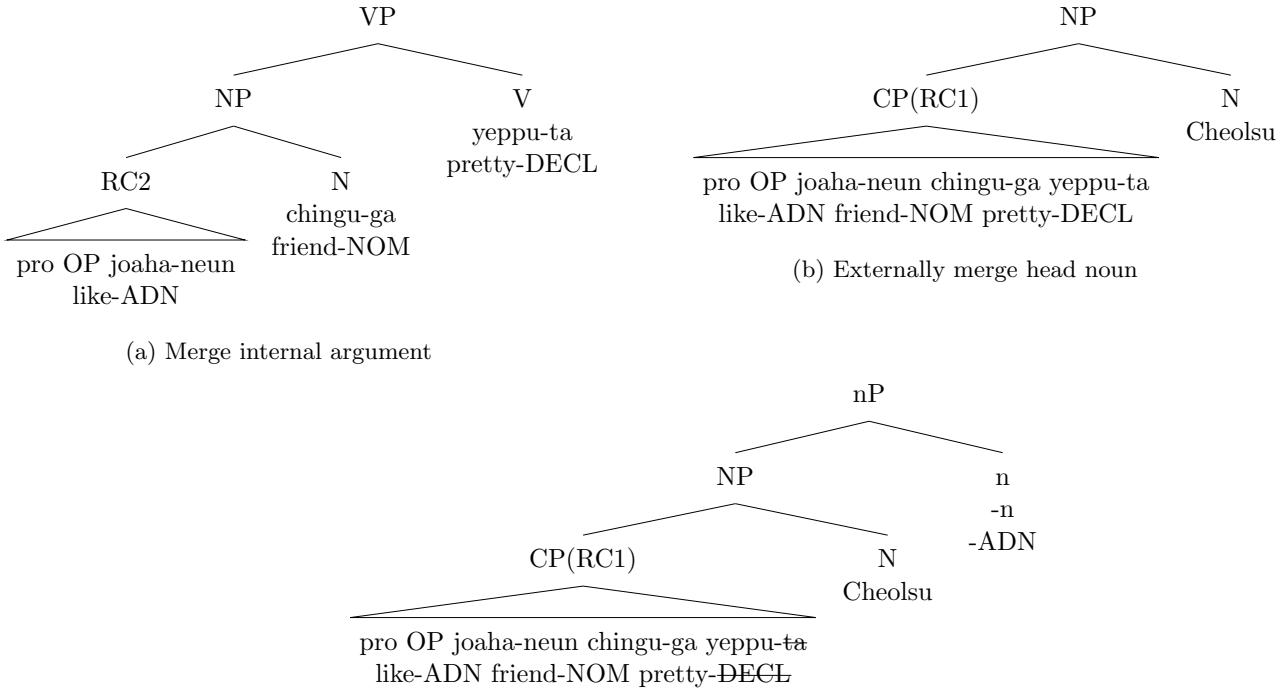
(c) Externally merge head noun

(d) Project little *n* and Merge the RC Marker

Figure 29: Formation of NP2 in the DRC Construction

As before, we defer an explanation of how the word order in Figure 29d is linearized from *neun chingu-ga joaha-* to *joaha-neun chingu-ga* to the following section. Thus far, we have exhausted all

lexical items except the adjective *yeppu-ta* 'pretty-DECL' from the numeration in N_2 of Figure 28, while two lexical items pertaining to the numeration of the matrix NP are all unused. Taking the linearized output of Figure 29 as a single object constituent, we can complete the derivation of the final relative clause—the outer RC1—modifying the first subject head noun.



(a) Merge internal argument

(b) Externally merge head noun

(c) Project little n and Merge the RC Marker

Figure 30: Relativization of the First Subject in Double Relative Clauses

However, as can be seen in Figure 30, the outer merging process does not involve merger of an external argument. This fact follows directly from the way the lexical items in the numeration are specified. Specifically, the gap corresponding to the first subject is instantiated as a null pronoun *pro*, whereas the gap associated with the second subject involves an operator (as shown in Figure 29b). As argued by Han and Kim (2004), this asymmetry entails that relativization of the first subject essentially involves no movement, unlike cases in which an operator is involved and must undergo movement to spec CP. As a result, no island violations arise, since there is no movement dependency to be constrained in the first place.

The distinct nature of these gaps has been consistently indicated throughout the derivation. In Figure 16, the first gap e_i is annotated as pro_i, while the second gap e_j is not. This asymmetry is reflected in the absence of movement of pro_i in RC1, as illustrated in Figure 18.

3.2 Addressing the Problem in Linearization

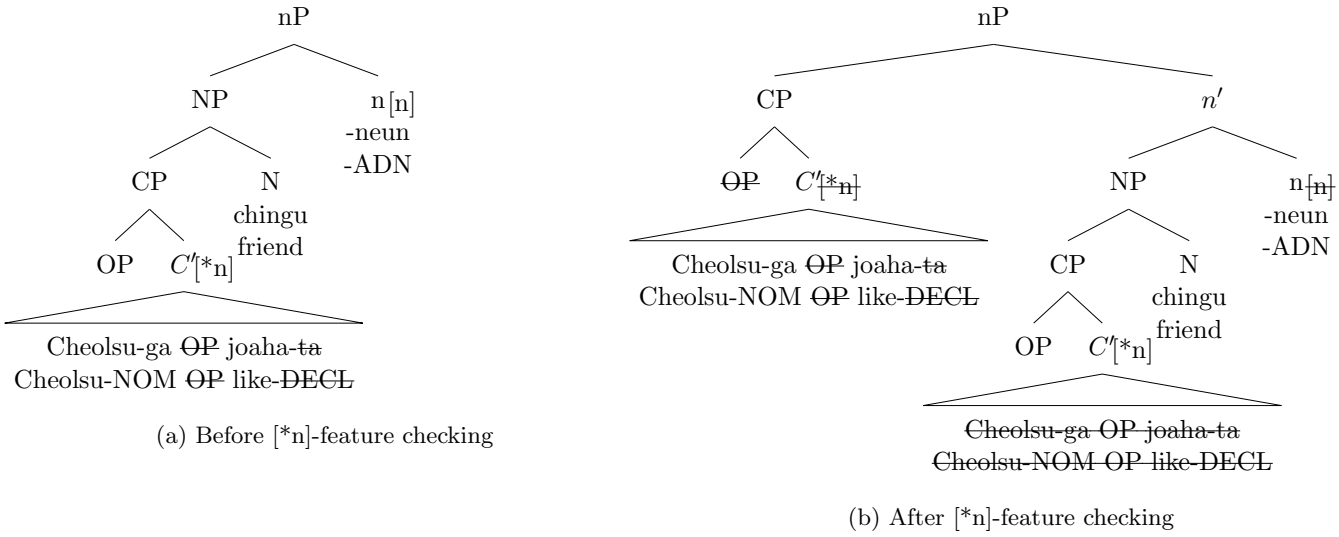
Finally, we return to the linearization issues left unaddressed in Figures 26d, 29d, and 30c. In accounting for how these structures ultimately yield the pre-nominal modifying word order, we

provide fully expanded trees involving operator movement, which underlies our proposal but was not fully reflected in the earlier diagrams.

In the overview of section 3, we hypothesized that Korean relative clauses bear a strong [n] feature, which triggers movement to spec nP so that the feature can be checked by the n head in a local spec-head configuration. This section is organized by tracing how each tree configuration triggers movement to satisfy this checking requirement.

We begin with the simplest case of single relative clause adjunction. Although omitted in the previous illustration (Figure 26), the operator moves out of its base-merged position prior to merger with the external head, i.e., before the C complement undergoes spell-out. This ordering was also taken to motivate why introducing relative clause morphology within TP would be inappropriate. Accordingly, Figure 26d is revised as Figure 31a, with operator movement to a spec CP position made explicit. We additionally mark a strong [n] feature on the relative clause at the C-bar level, notated as [*n], which triggers movement to spec nP in Figure 31a and yields the configuration in Figure 31b. In the latter, the CP containing the operator undergoes copy-and-merge, as the [*n] feature on the relative clause is checked against the [n] feature inherently present on little n.

Two points are noteworthy here. First, the operation targets the maximal projection CP containing the operator, in accordance with the Extension Condition, which requires that movement target only the root syntactic object (Hornstein et al., 2005). Second, it is the goal RC that bears the strong [*n] feature, rather than the probing n head itself. As discussed above, locating featural strength on the relative clause—rather than on the invariant projection head n—is more appropriate as the adjunct is optional and may not always be present to satisfy such a requirement.



(a) Before [*n]-feature checking

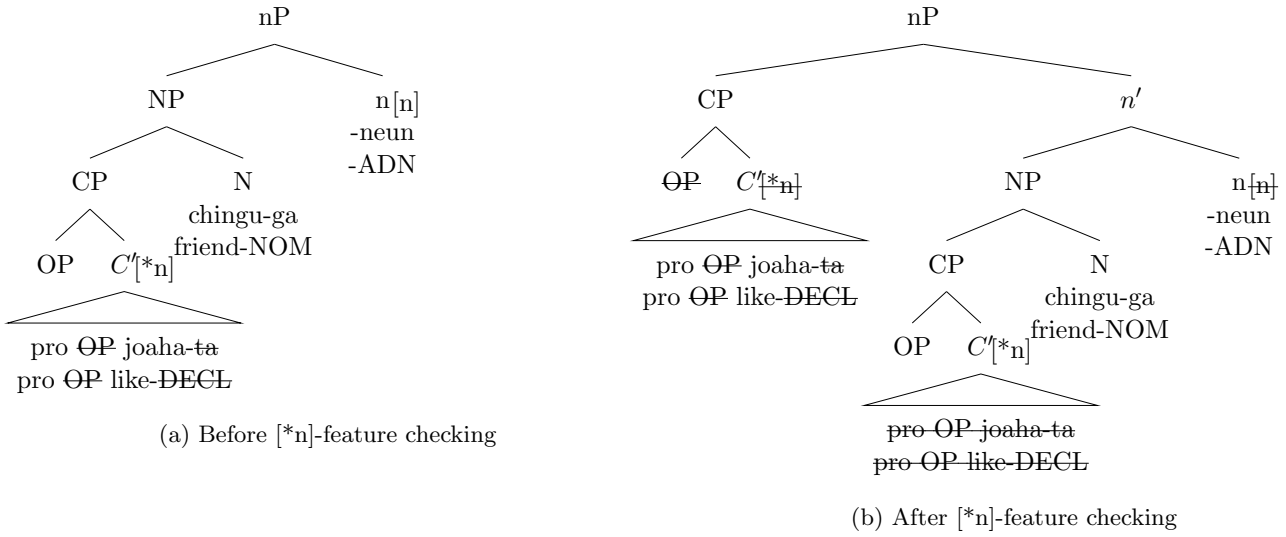
(b) After [*n]-feature checking

Figure 31: Linearization in Figure 26d

To avoid a contradiction at linearization—a configuration in which one element both asymmetrically

c-commands another and is simultaneously c-commanded by it—one of the copies created by copy-and-merge must be deleted. In the present case, deletion of the lower copy allows the relative clause to asymmetrically c-command the relative clause marker *-neun*, and subsequently the head noun, yielding the correct morphosyntactic configuration: *Cheolsu-ga joaha-neun chingu*.

Meanwhile, the operator is phonologically null and therefore irrelevant for computing precedence relations at the PF level. At LF, however, it is crucial for interpretation, as it occupies the verbal argument position within the relative clause. We propose that the operator undergoes scattered deletion at LF: in the higher copy, the phonologically overt relative clause is retained, while the operator is deleted; in the lower copy, the relative clause is deleted, and the operator is retained, leaving the operator stranded at the spec CP position. The stranded operator could be reanalyzed as the noun complement, allowing it to enter into a syntactic binding relation with the N head without recourse to semantic lambda abstraction, a point to which we return in section 3.3.



(a) Before [*n]-feature checking

(b) After [*n]-feature checking

Figure 32: Linearization in Figure 29d

The same derivational steps apply in deriving the correct word order in Figure 29d. In Figure 32, the operator first moves out of the relative clause, after which the CP containing the operator in its specifier is copied and merged to spec nP in order to satisfy the strong [*n] feature on the relative clause (C'). The only difference from the previous case is that the subject of the relative clause is a null pronoun, coindexed with the head noun with which the structure in Figure 32b subsequently merges. This configuration corresponds to the internally merged NP object in Figure 30a, which was previously simplified for representational purposes. To make explicit the connection between Figure 32b and Figure 30a, we redraw an expanded version of the tree in Figure 33.

Taking the entire structure in Figure 33 as a fully formed adjunct (RC1 in Figure 30b) ready to modify the first subject, *Cheolsu*, we now return to the question of how linearization proceeds in Figure 30c. Once again, we observe operator movement, alongside deletion of the declarative

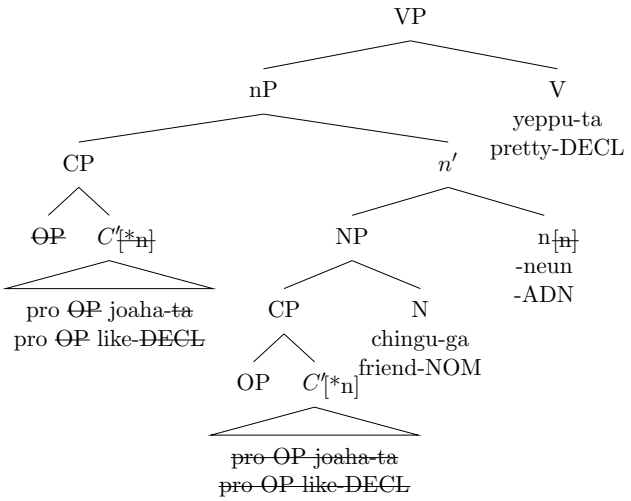
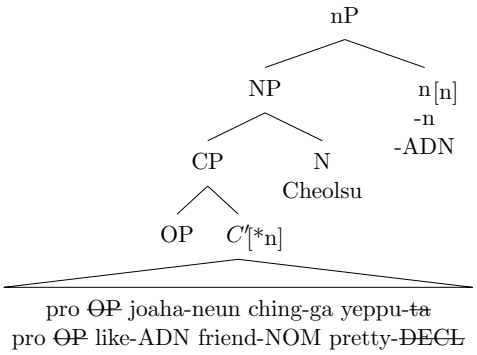
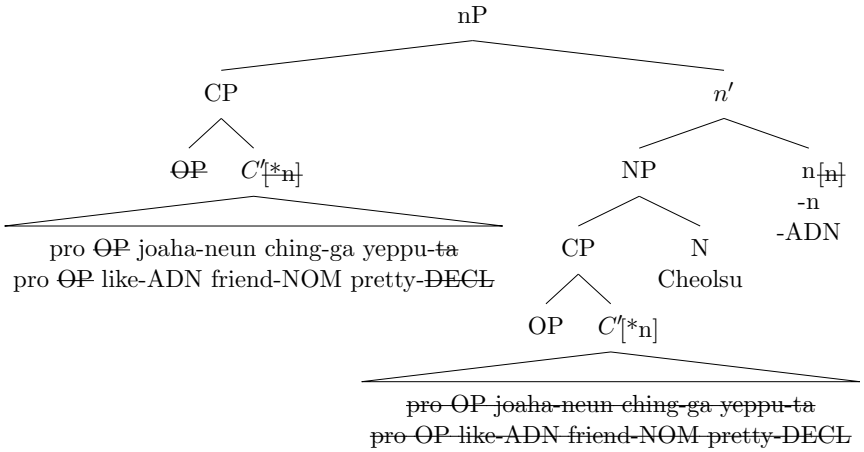


Figure 33: Expanded Tree of Figure 30a

ending to satisfy the PF constraint in Definition 1, at the initial stage before checking the [*n] feature (Figure 34a). Subsequently, in Figure 34b, the CP containing the relative clause moves to spec nP, satisfying [*n]. At PF, the lower copy of the relative clause is deleted. At LF, scattered deletion applies, such that the operator deletes in the higher copy and remains in the lower copy.



(a) Before [*n]-feature checking



(b) After [*n]-feature checking

Figure 34: Linearization in Figure 30c

3.3 Interpretive Advantages

One thing to note in the preceding Korean tree diagrams is the absence of a D projection. Figure 24b constitutes an exception, where we intentionally projected the demonstrative D to draw a parallel with the English example concerning the positionality of modifying phrases. Even in this case, however, Korean lacks grammatical articles and permits demonstratives only optionally. Ultimately,

a DP layer seems to lack an independent motivation in Korean nominal structures generally.

One context in which a DP projection becomes necessary is the establishment of possessive relations between two nominals, as in complex noun phrases of the form NP2's NP1. Unlike the Saxon genitive in English, however, the Korean genitive marker *-uy* cannot express an agent–action relation with an event-denoting noun, as in the example *Morticia's photograph of Jenny*, where Morticia is interpreted as the agent responsible for the photographing event. The direct translation *Morticia-uy Jenny-sajin* ('Morticia-GEN Jenny-picture') would only be interpreted as a possessive relation, namely that Morticia possesses a picture of Jenny. All constructions involving a D head projection, then, seem to correspond to what Adger (2003) terms independent genitives.

As can be seen from the example sentence above, in argumental nominal constructions, the possessor is an agent argument that bears possessive marking. Accordingly, such arguments are analyzed as being raised from a base-generated position in spec nP, as Figure 35a illustrates. By contrast, in independent genitives derived from a lexical N that does not license arguments, it is difficult to maintain that the possessor is likewise generated in spec nP. Since such possessors do not bear an agent theta-role, assigning them the same base position would violate the Uniformity of Theta Assignment Hypothesis (UTAH). Adger (2003) tentatively proposes the functional projection Poss, whose specifier can alternatively serve as the base-generated position for the possessor element (Figure 35b). Nevertheless, the theoretical mechanisms responsible for licensing independent genitives remain under debate.

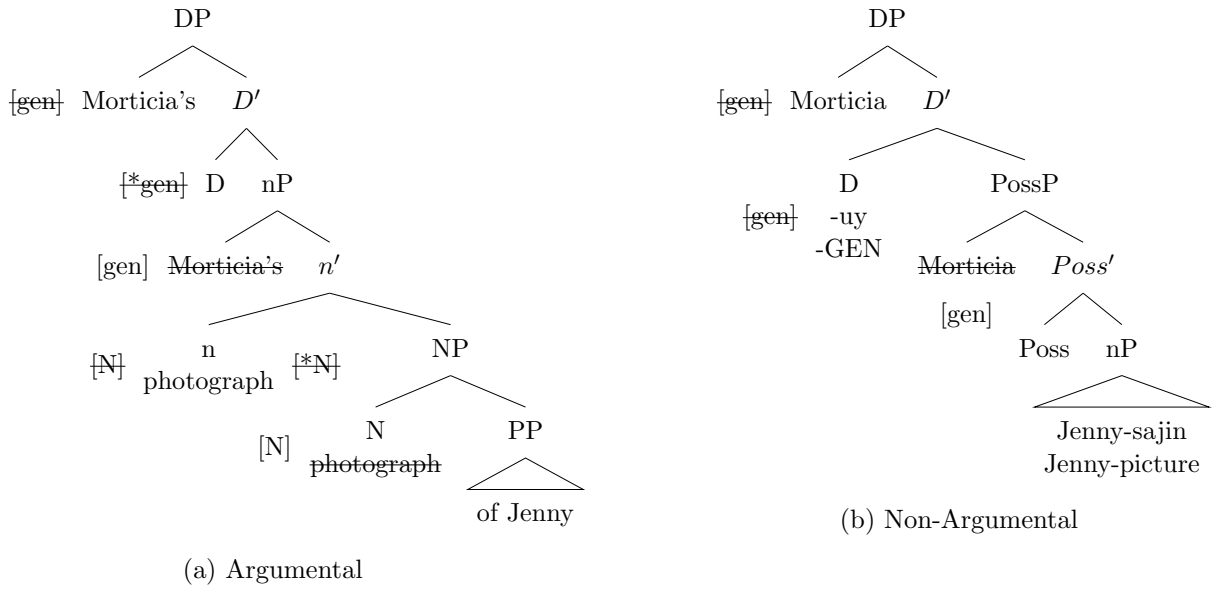
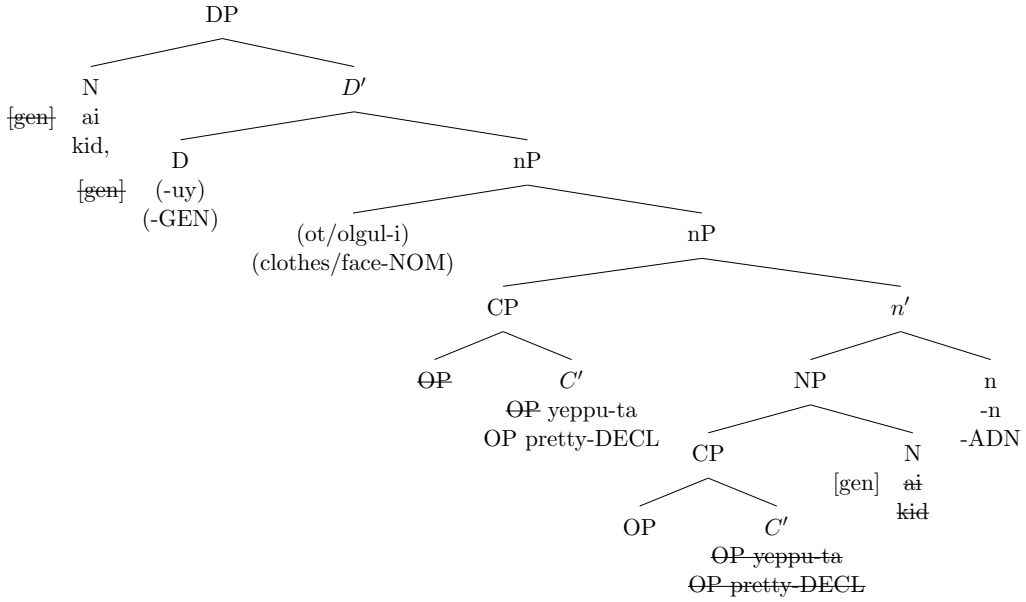


Figure 35: Possessors in Argumental vs. Non-Argumental Nominal Constructions

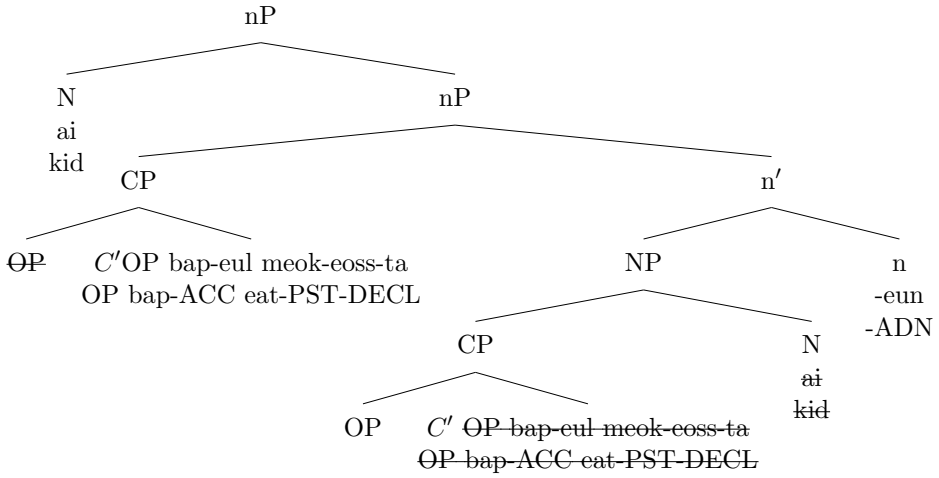
Our analysis of relative clauses in the nominal domain contributes intriguing empirical data to this ongoing debate. In the interpretation of stand-alone relative clauses (Figure 5), the head noun is construed as the possessor and undergoes movement to spec DP. This derivation is made explicit

in the reconstruction tree in Figure 6, which illustrates how the de-relativization stage proceeds at LF.

As the head noun stands in a possessive relationship to the elided trait subject (height/size/head), it is interpreted as an element bearing genitive case, against which the D head can check its case feature. Consequently, it is neither the constituent base-generated in spec nP nor the constituent base-generated in the specifier of the alternative functional head Poss that raises to spec DP, but rather the head noun itself, which projects the entire nominal domain.



(a) stand-alone type



(b) phrase type (subject gap)

Figure 36: Reconstruction of Stand-alone and Phrase Types

This contrasts with the reconstruction of the phrase-type, which merely involves raising the head noun to the gapped position where it receives its interpretative theta role. Since our focus is on

stand-alone relative clauses, whose syntactic gap is limited to the subject position, we compare the reconstruction tree of the stand-alone possessive interpretation of *[yeppeu-n] ai* (pretty-ADN kid) with its phrasal-construction counterpart involving a subject gap, *[bap-eul meok-eun] ai* (meal-ACC eat-ADN kid).

The interpretive theta role of the subject gap is agent, paralleling the clausal domain, where the verbal argument bearing the agent role checks nominative case. In the reconstructed sentence in Figure 36b, *ai-(ga) bap-eul meok-eoss-ta* (kid-(NOM) meal-ACC eat-PST-DECL), the head noun kid is the agent of the action of the eating event and therefore raises to the spec nP, which is analogous to how it would have been base-generated at the spec vP in the clausal domain. Notice that, since we are accounting for the operations at LF, the PF ending morpheme deletion rule in Definition 1 does not apply in either tree, as the declarative suffix *-ta* and the adnominal suffix *-eun* coexist. Regardless, the linear precedence relation between these two morphemes should not interfere with deriving the correct interpretation.

Additionally, in Figure 36b, the past tense morpheme *-eoss* is expected to be deleted at PF by a constraint prohibiting the co-occurrence of *-eoss* with the relative clause marker *-eun* (Hong, 2012).

Importantly, the head noun *kid* in Figure 36a bears weak genitive case and raises to spec DP at LF to check this feature covertly, thereby establishing a possessive relation with the elided subject in the NP2's NP1 structure discussed above. As noted, this is the only construction that requires a projection of the D head. In this configuration, the elided subject is generated in spec nP in accordance with UTAH, since this is the position in which the Agent theta role is assigned. This generation is not phonologically overt and therefore proceeds covertly; moreover, the elided subject is not triggered to move to spec DP, as it is the head noun possessor that checks the [gen] feature on D. This structural movement contrasts with Figure 36b, where the same head noun raises to spec nP in order to receive its interpretive theta role.

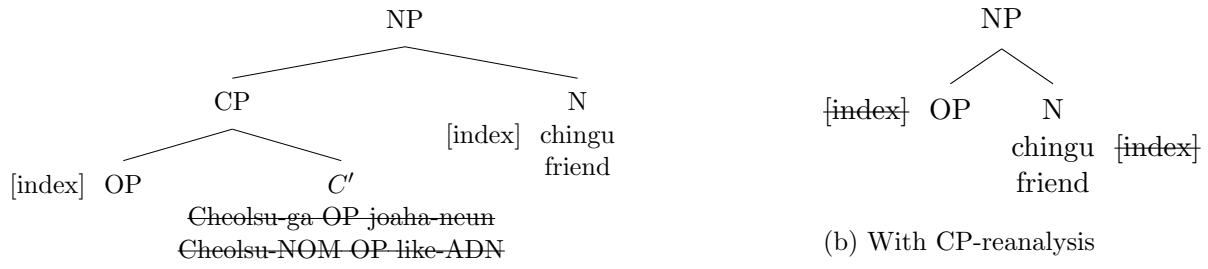
Ultimately, this comparison, on the one hand, provides further insight into the theory of nominal domain by suggesting an alternative position from which possessors of independent genitive construction may originate. On the other hand, adopting a theoretical framework that posits projections of functional categories beyond the lexical NP domain offers an interpretive advantage in structurally accounting not only for the derivation and linearization of phrase-type relative clauses, but also for the establishment of possessive interpretations in stand-alone relative clauses, given the optional projection of the D head in Korean.

Before concluding, we must address the implications of scattered deletion at LF between the operator and the phonologically overt relative clause. As consistently noted throughout the paper, the stranded operator at the lower copy can be reanalyzed as an independent argument that is eligible for syntactic licensing in a local head–complement configuration.

The notion of reanalysis adopted here is inspired by the idea of v'-reanalysis proposed in Larson (1988), according to which a verb–goal constituent within the VP-internal hierarchy may be reanalyzed as a head that is eligible to raise to an empty head position, in a process known as light predicate raising. The motivation for treating the v' constituent as an argument capable of head-to-head movement lies in the presence of an unsaturated theta role within the phrase. We extend this logic to the nominal domain by formulating a rule under which a CP bearing an unchecked [index] feature may be reanalyzed as an independent argument, as stated in definition 2.

Definition 2 (CP-reanalysis)**.** *Let α be a phrase [CP...] whose feature bundle contains one unchecked feature. Then α may be reanalyzed as [C...].*

Let us further assume that, in order for the operator to be coindexed with the head noun it refers to, its uninterpretable [index] feature must be checked. Under standard assumptions in Minimalism, agreement relations are established only in local configurations, namely spec–head or head–complement configurations. Prior to CP reanalysis, the operator is embedded too deeply within the structure to enter into an agree relation with the head noun, and thus cannot have its [index] feature checked. Consequently, the configuration fails to meet the requirements for syntactic licensing, as illustrated in Figure 37a.We use the earlier example in Figure 8b with scattered deletion applied, as the diagram represents the configuration at the lower copy.



(a) Without CP-reanalysis

(b) With CP-reanalysis

Figure 37: Co-indexing Agreement

We schematize the reanalysis as the removal of the CP skeleton surmounting the operator *OP*. In Figure 37b, only a single operator argument is left in the place previously adjoined by the relative clause phrase. Consequently, the operator and the head noun can enter into an agree relation and check their index features, thereby achieving a syntactic licensing.

In the end, our proposal—which relocates the CP containing the relative clause to spec nP—derives not only the correct pre-nominal word order but also generates two copies of the clausal phrase containing the relative clause. While adopting the copy theory of movement, the higher copy, created in the movement-induced merged position, bears the burden of realizing the phonologically overt relative clause, while at the base-generated position, only an operator may remain in the specifier position of the lower copy. This operator can subsequently be reanalyzed as a single argument in

a sisterhood relation with the head noun to which the CP initially adjoins. As a result, the operator can be licensed within the syntactic domain, without recourse to syntax-external semantic mechanisms.

References

Adger, D. (2003). *Core syntax: A minimalist approach.* Oxford University Press.

Bowers, J. (1993). The syntax of predication. *Linguistic inquiry*, 24(4):591–656.

Chae, H.-R. (2012). Are there relative clauses in korean? a participle clause analysis. *Korean Journal of Linguistics*, 37(4):1043–1065.

Chomsky, N. (2015). *The Minimalist Program.* The MIT Press, 20th edition.

Chomsky, N. and Kenstowicz, M. (1999). Derivation by phase. *An Annotated Syntax Reader*, 482.

Han, C.-h. (2013). On the syntax of relative clauses in korean. *Canadian Journal of Linguistics/Revue canadienne de linguistique*, 58(2):319–347.

Han, C.-h. and Kim, J.-B. (2004). Are there" double relative clauses" in korean? *Linguistic Inquiry*, pages 315–337.

Hong, Y. (2012). Relative clause markers in korean. *Korean Journal of Linguistics*, 37(4):1067–1094.

Hornstein, N., Nunes, J., and Grohmann, K. K. (2005). *Understanding minimalism.* Cambridge University Press.

Kang, M.-Y. (1988). Syntactic movement in korean relativization. *Linguistics in the morning calm*, 2:347–362.

Kayne, R. S. (1994). *The antisymmetry of syntax*, volume 25. MIT press.

Kwon, N. (2008). *Processing of syntactic and anaphoric gap-filler dependencies in Korean: Evidence from self-paced reading time, ERP and eye-tracking experiments.* University of California, San Diego.

Larson, R. K. (1988). On the double object construction. *Linguistic inquiry*, 19(3):335–391.

YOON, J. S. (1986). Some queries concerning the syntax of multiple subject constructions in korean in illinois studies in korean linguistics. *Studies in the Linguistic Sciences*, 16(2):215–236.